Complex Analysis II

Lecture notes following Stein–Shakarchi, Ch. 8–10 and Appendices A–B

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Chapter 8. Conformal Mappings

Overview

This chapter is geometric and "global": we study mapping properties of holomorphic functions, motivated by the question

> *Given two open sets U and V in $\mathbb{C}$, does there exist a holomorphic bijection between them?*

(The inverse is automatically holomorphic.) The key case is $V = \mathbb{D}$: *when is there a holomorphic bijection $\Omega \to \mathbb{D}$?* (For $V = \mathbb{C}$ only $U = \mathbb{C}$ works.) We give explicit maps and apply them to the Dirichlet problem, determine the automorphisms of $\mathbb{D}$ and $\mathbb{H}$ by the Schwarz lemma, and prove the *Riemann mapping theorem*: every proper simply connected Ω is conformally equivalent to $\mathbb{D}$, with no assumption on $\partial\Omega$. For polygons the map is the *Schwarz–Christoffel formula*; for rectangles it involves elliptic integrals, leading to the doubly periodic functions of Chapter 9.

1 Conformal equivalence and examples

Motivation. Regions are "the same" when a holomorphic bijection maps one onto the other; such maps transport problems to the disc or half-plane.

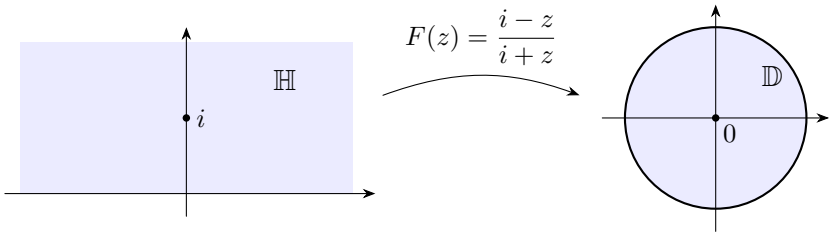


Figure. *The Cayley-type map $F\colon \mathbb{H} \to \mathbb{D}$, real axis onto the circle minus -1, $i \mapsto 0$.*

A bijective holomorphic $f : U \to V$ is a **conformal map** or **biholomorphism**; then U and V are **conformally equivalent** (**biholomorphic**).

Proposition 1.1

If $f : U \to V$ is holomorphic and injective, then $f'(z) \neq 0$ on U. So the inverse of f on its range, in particular of a conformal map, is holomorphic.

Proof. Suppose $f'(z_0) = 0$. Near z_0, $f(z) - f(z_0) = a(z - z_0)^k + G(z)$, $a \neq 0$, $k \geq 2$, G vanishing to order $k + 1$. For small w, $f(z) - f(z_0) - w = F(z) + G(z)$ with $F(z) = a(z - z_0)^k - w$; on a small circle $|G| < |F|$, and F has at least two zeros inside, so Rouché gives $f - f(z_0) - w$ at least two zeros, distinct as $f' \neq 0$ near z_0 ($z \neq z_0$). So f is not injective.

Let $g = f^{-1}$ on the range V, $w_0 = f(z_0)$, $w = f(z)$ close to w_0, $w \neq w_0$. Then

$$\frac{g(w) - g(w_0)}{w - w_0} = \frac{1}{\frac{w-w_0}{g(w)-g(w_0)}} = \frac{1}{\frac{f(z)-f(z_0)}{z-z_0}}.$$

Since $f'(z_0) \neq 0$, letting $z \to z_0$ shows g is holomorphic at w_0 with $g'(w_0) = 1/f'(g(w_0))$. ∎

So U, V are conformally equivalent iff there are holomorphic $f : U \to V$, $g : V \to U$ inverse to each other. (Some authors call f conformal when merely $f' \neq 0$, e.g. z^2 on $\mathbb{C} - \{0\}$; such f is a local bijection preserving angles.)

1.1 The disc and upper half-plane

The **upper half-plane** is $\mathbb{H} = \{z \in \mathbb{C} : \mathrm{Im}(z) > 0\}$. Let

$$F(z) = \frac{i - z}{i + z} \qquad \text{and} \qquad G(w) = i\frac{1 - w}{1 + w}.$$

Theorem 1.2

The map $F : \mathbb{H} \to \mathbb{D}$ is a conformal map with inverse $G : \mathbb{D} \to \mathbb{H}$.

Proof. Both maps are holomorphic in their domains. Points of $\mathbb{H}$ are closer to i than to $-i$, so $|F(z)| < 1$. For $w = u + iv \in \mathbb{D}$,

$$\mathrm{Im}(G(w)) = \mathrm{Re}\left(\frac{1 - u - iv}{1 + u + iv}\right) = \mathrm{Re}\,\frac{(1 - u - iv)(1 + u - iv)}{(1 + u)^2 + v^2} = \frac{1 - u^2 - v^2}{(1 + u)^2 + v^2} > 0.$$

Finally

$$F(G(w)) = \frac{i - i\frac{1-w}{1+w}}{i + i\frac{1-w}{1+w}} = \frac{1 + w - 1 + w}{1 + w + 1 - w} = w,$$

and similarly $G(F(z)) = z$. ∎

F is holomorphic on $\mathbb{C}$ except at $-i$, hence continuous on $\mathbb{R} = \partial\mathbb{H}$, and $|F(x)| = 1$ for real x. With $x = \tan t$, $t \in (-\pi/2, \pi/2)$,

$$F(x) = \frac{1 - x^2}{1 + x^2} + i\frac{2x}{1 + x^2} = \cos 2t + i \sin 2t = e^{2it},$$

so F maps $\mathbb{R}$ onto the circle minus -1, traversed from -1 first through the lower half; -1 corresponds to the point at infinity of $\mathbb{H}$.

Remark

Maps $z \mapsto \frac{az+b}{cz+d}$ (denominator not a multiple of the numerator) are **fractional linear transformations**; they give the automorphisms of $\mathbb{D}$ and $\mathbb{H}$ (Theorems 2.2, 2.4).

1.2 Further examples

Example 1

Translations and dilations. $z \mapsto z + h$ is conformal on $\mathbb{C}$ (and on $\mathbb{H}$ for real h); $z \mapsto cz$, $c \neq 0$, is conformal on $\mathbb{C}$: a rotation for $c = e^{i\varphi}$, a dilation for $c > 0$, a dilation by $|c|$ followed by rotation by π for $c < 0$.

Example 2

$z \mapsto z^n$ maps $\{0 < \arg z < \pi/n\}$ conformally onto $\mathbb{H}$ (inverse $w^{1/n}$, principal branch). For $0 < \alpha < 2$, $f(z) = z^\alpha$ maps $\mathbb{H}$ onto $S = \{0 < \arg w < \alpha\pi\}$: deleting the positive real axis, $f(re^{i\theta}) = r^\alpha e^{i\alpha\theta}$, $0 < \theta < \pi$, with inverse $w^{1/\alpha}$. With Example 1, $\mathbb{H}$ maps onto any infinite sector. On the boundary, $(-\infty, 0]$ goes onto the ray $\arg z = \alpha\pi$ (from $\infty e^{i\alpha\pi}$ to 0) and $[0, \infty)$ onto $[0, \infty)$.

Example 3

$f(z) = \frac{1+z}{1-z}$ maps the upper half-disc $\{|z| < 1,\ y > 0\}$ conformally onto the first quadrant $\{u > 0, v > 0\}$: for $z = x + iy$,

$$f(z) = \frac{1-(x^2+y^2)}{(1-x)^2+y^2} + i\frac{2y}{(1-x)^2+y^2}.$$

The inverse $g(w) = \frac{w-1}{w+1}$ maps the first quadrant into $\mathbb{D}$ (w is nearer 1 than -1) with $\operatorname{Im} g > 0$. On the boundary, $f(e^{i\theta}) = \frac{e^{-i\theta/2}+e^{i\theta/2}}{e^{-i\theta/2}-e^{i\theta/2}} = \frac{i}{\tan(\theta/2)}$ runs down the imaginary axis from ∞ to 0 as θ goes from 0 to π; f maps $(-1,1)$ increasingly onto $(0,\infty)$, $f(0) = 1$.

Example 4

$\log z$ (branch deleting the negative imaginary axis, $\log(re^{i\theta}) = \log r + i\theta$, $-\pi/2 < \theta < 3\pi/2$) maps $\mathbb{H}$ onto $\{0 < v < \pi\}$, inverse e^w; $(-\infty, 0)$ goes to the line $v = \pi$ (from $\infty + i\pi$ to $-\infty + i\pi$), $(0,\infty)$ onto $\mathbb{R}$.

Example 5

$\log z$ also maps the half-disc $\{|z| < 1, y > 0\}$ onto $\{u < 0,\ 0 < v < \pi\}$: $(0,1) \to (-\infty, 0)$, the upper half-circle to $[0, i\pi]$, $(-1,0)$ to the half-line from $i\pi$ to $-\infty + i\pi$.

Example 6

$f(z) = e^{iz} = e^{-y}e^{ix}$ maps $\{-\pi/2 < x < \pi/2,\ y > 0\}$ onto $\{|w| < 1,\ u > 0\}$. Boundary: the right side goes from 0 to i, $[-\pi/2, \pi/2]$ along the half-circle from i to $-i$, the left side from $-i$ to 0. (Cf. the inverse of Example 5.)

Example 7

$f(z) = -\frac{1}{2}(z + 1/z)$ maps $\{|z| < 1, y > 0\}$ conformally onto $\mathbb{H}$. Boundary: $(0,1) \to (-\infty, -1)$; on $z = e^{i\theta}$, $f = -\cos\theta$ runs from -1 to 1; $(-1,0) \to (1,\infty)$.

Example 8

$f(z) = \sin z$ maps the half-strip $\{-\pi/2 < x < \pi/2,\ y > 0\}$ conformally onto $\mathbb{H}$. With $\zeta = e^{iz}$,

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i} = -\frac{1}{2}\Big(i\zeta + \frac{1}{i\zeta}\Big),$$

so f is Example 6, then multiplication by i, then Example 7. The left side goes to $(-\infty, -1]$, $[-\pi/2, \pi/2]$ onto $[-1, 1]$, the right side to $[1,\infty)$.

1.3 The Dirichlet problem in a strip

The **Dirichlet problem** in Ω is

$$\triangle u = 0 \text{ in } \Omega, \qquad u = f \text{ on } \partial\Omega, \tag{1}$$

$\triangle = \partial^2/\partial x^2 + \partial^2/\partial y^2$, f given on $\partial\Omega$. Poisson kernels solve it for the disc and $\mathbb{H}$; conformal maps give the strip.

Lemma 1.3

Let V and U be open sets in $\mathbb{C}$ and $F : V \to U$ a holomorphic function. If $u : U \to \mathbb{C}$ is a harmonic function, then $u \circ F$ is harmonic on V.

Proof. The question is local; on a disc U, $u = \operatorname{Re} G$ with G holomorphic, so $u \circ F = \operatorname{Re}(G \circ F)$ is harmonic. ∎

Let $\Omega = \{x + iy : x \in \mathbb{R},\ 0 < y < 1\}$, with boundary data $u(x, 0) = f_0(x)$, $u(x, 1) = f_1(x)$, f_0, f_1 continuous and vanishing at infinity. Transfer the problem to $\mathbb{D}$ by

$$F(w) = \frac{1}{\pi} \log\Big(i\frac{1-w}{1+w}\Big) : \mathbb{D} \to \Omega, \qquad G(z) = \frac{i - e^{\pi z}}{i + e^{\pi z}} : \Omega \to \mathbb{D},$$

compositions of maps above, mutually inverse. F maps the lower half-circle onto $i + \mathbb{R}$ (from $i + \infty$ to $i - \infty$) and the upper one onto $\mathbb{R}$ (from $-\infty$ to ∞). Let $\tilde{f}_1(\varphi) = f_1(F(e^{i\varphi}) - i)$, $-\pi < \varphi < 0$, and $\tilde{f}_0(\varphi) = f_0(F(e^{i\varphi}))$, $0 < \varphi < \pi$; as f_0, f_1 vanish at infinity, $\tilde{f}$ (0 at $\varphi = \pm\pi, 0$) is continuous on the circle. The disc solution is the Poisson integral, for $w = re^{i\theta}$,

$$\tilde{u}(w) = \frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(\theta - \varphi)\tilde{f}(\varphi)\, d\varphi = \frac{1}{2\pi}\int_{-\pi}^{0} P_r(\theta - \varphi)\tilde{f}_1(\varphi)\, d\varphi + \frac{1}{2\pi}\int_{0}^{\pi} P_r(\theta - \varphi)\tilde{f}_0(\varphi)\, d\varphi,$$

$P_r(\gamma) = \frac{1-r^2}{1-2r\cos\gamma+r^2}$ the Poisson kernel. By Lemma 1.3, $u(z) = \tilde{u}(G(z))$ is harmonic in the strip, with the correct boundary values. For $z = iy$, $0 < y < 1$, with $re^{i\theta} = G(iy)$, a change of variables gives

$$\frac{1}{2\pi}\int_{0}^{\pi} P_r(\theta-\varphi)\tilde{f}_0(\varphi)\, d\varphi = \frac{\sin \pi y}{2}\int_{-\infty}^{\infty} \frac{f_0(t)}{\cosh \pi t - \cos \pi y}\, dt, \qquad \frac{1}{2\pi}\int_{-\pi}^{0} P_r(\theta-\varphi)\tilde{f}_1(\varphi)\, d\varphi = \frac{\sin \pi y}{2}\int_{-\infty}^{\infty} \frac{f_1(t}{\cosh \pi t +}$$

A solution vanishing at infinity is unique, so translating the boundary data by x translates the solution, and

$$u(x, y) = \frac{\sin \pi y}{2}\Big(\int_{-\infty}^{\infty} \frac{f_0(x-t)}{\cosh \pi t - \cos \pi y}\, dt + \int_{-\infty}^{\infty} \frac{f_1(x-t)}{\cosh \pi t + \cos \pi y}\, dt\Big),$$

convolutions with f_0 and f_1. At $y = 1/2$ the kernel is $1/\cosh \pi t$, its own Fourier transform (Example 3 in Section 2 of Chapter 3).

Remarks about the Dirichlet problem. If $F : \mathbb{D} \to \Omega$ is conformal and extends to a continuous bijection $\partial\mathbb{D} \to \partial\Omega$, then with $\tilde{f} = f \circ F$ on the circle and $\tilde{u}(re^{i\theta}) = \frac{1}{2\pi}\int_0^{2\pi} P_r(\theta - \varphi)\tilde{f}(e^{i\varphi})\, d\varphi$, $u = \tilde{u} \circ F^{-1}$ should solve (1). This requires:

(i) existence of a conformal map $\Phi = F^{-1} : \Omega \to \mathbb{D}$, settled in general by the Riemann mapping theorem;

(ii) its extension to a continuous bijection of the boundaries, needing some regularity of $\partial\Omega$ (polygons: Section 4). Riemann reversed this: writing $\Phi(z) = (z - z_0)e^{H(z)}$, $|\Phi| = 1$ on $\partial\Omega$ means $u = \operatorname{Re} H$ solves the Dirichlet problem with data $\log(1/|z - z_0|)$ (the Green's function). This needs bijectivity of Φ, boundary regularity, and a solution of the Dirichlet problem (Riemann's "Dirichlet principle" is delicate). Section 3 proves existence differently.

2 The Schwarz lemma; automorphisms of the disc and upper half-plane

Motivation. The Schwarz lemma forces the automorphisms of $\mathbb{D}$ and $\mathbb{H}$ to be Möbius transformations.

A **rotation** is a map $z \mapsto cz$ with $|c| = 1$, $c = e^{i\varphi}$, φ the angle of rotation (defined modulo 2π).

Lemma 2.1 (Schwarz lemma)

Let $f : \mathbb{D} \to \mathbb{D}$ be holomorphic with $f(0) = 0$. Then (i) $|f(z)| \le |z|$ for all $z \in \mathbb{D}$.
(ii) If for some $z_0 \neq 0$ we have $|f(z_0)| = |z_0|$, then f is a rotation.
(iii) $|f'(0)| \le 1$, and if equality holds, then f is a rotation.

Proof. $f(0) = 0$, so $f(z)/z$ is holomorphic in $\mathbb{D}$. On $|z| = r < 1$, $|f(z)/z| \le 1/r$, hence on $|z| \le r$ by the maximum modulus principle; $r \to 1$ gives (i). For (ii), $f(z)/z$ attains its maximum inside $\mathbb{D}$, so it is a constant c, $|c| = 1$. For (iii), $g = f(z)/z$ has $|g| \le 1$ and $g(0) = f'(0)$; if $|f'(0)| = 1$, g is constant by the maximum principle, $f(z) = cz$, $|c| = 1$. ■

2.1 Automorphisms of the disc

A conformal map of Ω onto itself is an **automorphism**; they form the group $\mathrm{Aut}(\Omega)$. Rotations $r_\theta : z \mapsto e^{i\theta} z$ are automorphisms of $\mathbb{D}$. More interesting are

$$\psi_\alpha(z) = \frac{\alpha - z}{1 - \overline{\alpha} z}, \qquad \alpha \in \mathbb{C},\ |\alpha| < 1.$$

ψ_α is holomorphic in $\mathbb{D}$. If $z = e^{i\theta}$, $\psi_\alpha(e^{i\theta}) = \frac{\alpha - e^{i\theta}}{e^{i\theta}(e^{-i\theta} - \overline{\alpha})} = -e^{-i\theta}\frac{w}{\overline{w}}$ with $w = \alpha - e^{i\theta}$, so $|\psi_\alpha| = 1$ on the circle and $|\psi_\alpha| < 1$ in $\mathbb{D}$ by the maximum modulus principle. Moreover

$$(\psi_\alpha \circ \psi_\alpha)(z) = \frac{\alpha - \frac{\alpha - z}{1 - \overline{\alpha} z}}{1 - \overline{\alpha}\frac{\alpha - z}{1 - \overline{\alpha} z}} = \frac{\alpha - |\alpha|^2 z - \alpha + z}{1 - \overline{\alpha} z - |\alpha|^2 + \overline{\alpha} z} = \frac{(1 - |\alpha|^2) z}{1 - |\alpha|^2} = z,$$

so ψ_α is its own inverse; and it interchanges 0 and α: $\psi_\alpha(0) = \alpha$, $\psi_\alpha(\alpha) = 0$.

Theorem 2.2

If f is an automorphism of the disc, then there exist $\theta \in \mathbb{R}$ and $\alpha \in \mathbb{D}$ such that

$$f(z) = e^{i\theta} \frac{\alpha - z}{1 - \overline{\alpha} z}.$$

Proof. There is a unique $\alpha \in \mathbb{D}$ with $f(\alpha) = 0$. Let $g = f \circ \psi_\alpha$; then $g(0) = 0$ and by the Schwarz lemma

$$|g(z)| \le |z| \qquad \text{for all } z \in \mathbb{D}. \tag{2}$$

Also $g^{-1}(0) = 0$, so $|g^{-1}(w)| \le |w|$; with $w = g(z)$,

$$|z| \le |g(z)| \qquad \text{for all } z \in \mathbb{D}. \tag{3}$$

By (2), (3), $|g(z)| = |z|$, so $g(z) = e^{i\theta} z$ by the Schwarz lemma. Replacing z by $\psi_\alpha(z)$ and using $\psi_\alpha \circ \psi_\alpha = \mathrm{id}$, $f(z) = e^{i\theta}\psi_\alpha(z)$. ■

Corollary 2.3

The only automorphisms of the unit disc that fix the origin are the rotations.

$\mathrm{Aut}(\mathbb{D})$ acts *transitively* ($\psi_\beta \circ \psi_\alpha$ maps α to β) and is "almost" isomorphic to $SU(1,1)$, the 2×2 complex matrices preserving $\langle Z, W\rangle = z_1\overline{w_1} - z_2\overline{w_2}$.

2.2 Automorphisms of the upper half-plane

With $F : \mathbb{H} \to \mathbb{D}$ from Section 1, define $\Gamma : \mathrm{Aut}(\mathbb{D}) \to \mathrm{Aut}(\mathbb{H})$ by "conjugation by F": $\Gamma(\varphi) = F^{-1} \circ \varphi \circ F$. Γ is a bijection with inverse $\Gamma^{-1}(\psi) = F \circ \psi \circ F^{-1}$, and it preserves composition:

$$\Gamma(\varphi_1 \circ \varphi_2) = F^{-1} \circ \varphi_1 \circ F \circ F^{-1} \circ \varphi_2 \circ F = \Gamma(\varphi_1) \circ \Gamma(\varphi_2).$$

So $\mathrm{Aut}(\mathbb{D}) \cong \mathrm{Aut}(\mathbb{H})$. Let $SL_2(\mathbb{R}) = \left\{ M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{R},\ \det(M) = ad - bc = 1 \right\}$, the **special linear group**, and for $M \in SL_2(\mathbb{R})$ let $f_M(z) = \frac{az+b}{cz+d}$.

Theorem 2.4

Every automorphism of $\mathbb{H}$ takes the form f_M for some $M \in SL_2(\mathbb{R})$. Conversely, every map of this form is an automorphism of $\mathbb{H}$.

Proof. Write $G = SL_2(\mathbb{R})$.

Step 1. If $M \in G$, f_M maps $\mathbb{H}$ to itself:

$$\mathrm{Im}(f_M(z)) = \frac{(ad - bc)\,\mathrm{Im}(z)}{|cz + d|^2} = \frac{\mathrm{Im}(z)}{|cz + d|^2} > 0 \qquad (z \in \mathbb{H}). \tag{4}$$

Step 2. $f_M \circ f_{M'} = f_{MM'}$ (a straightforward calculation). Hence each f_M is an automorphism, with holomorphic inverse $f_{M^{-1}}$: $(f_M \circ f_{M^{-1}})(z) = f_I(z) = z$.

Step 3. G acts transitively on $\mathbb{H}$: it suffices to map $z \in \mathbb{H}$ to i. With $d = 0$ in (4), $\mathrm{Im}(f_M(z)) = \mathrm{Im}(z)/|cz|^2$; choose $c \in \mathbb{R}$ making this 1, $M_1 = \begin{pmatrix} 0 & -c^{-1} \\ c & 0 \end{pmatrix}$, then translate by $M_2 = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$, $b \in \mathbb{R}$; $M = M_2 M_1$.

Step 4. For $\theta \in \mathbb{R}$, $M_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \in G$, and $F \circ f_{M_\theta} \circ F^{-1}$ is the rotation of angle -2θ in the disc, since $F \circ f_{M_\theta} = e^{-2i\theta} F(z)$ (easily verified).

Step 5. Let $f \in \mathrm{Aut}(\mathbb{H})$ with $f(\beta) = i$, and $N \in G$ with $f_N(i) = \beta$. Then $g = f \circ f_N$ fixes i, so $F \circ g \circ F^{-1}$ is an automorphism of $\mathbb{D}$ fixing 0, hence a rotation; by Step 4, $F \circ g \circ F^{-1} = F \circ f_{M_\theta} \circ F^{-1}$ for some θ. So $g = f_{M_\theta}$ and $f = f_{M_\theta N^{-1}}$. ∎

M and $-M$ give the same f_M; identifying them gives the **projective special linear group** $PSL_2(\mathbb{R})$, which is isomorphic to $\mathrm{Aut}(\mathbb{H})$.

3 The Riemann mapping theorem

3.1 Necessary conditions and statement of the theorem

For $\Omega = \mathbb{C}$ there is no conformal map to $\mathbb{D}$ (Liouville); Ω must be connected and simply connected, like $\mathbb{D}$. These conditions suffice. A subset of $\mathbb{C}$ is **proper** if it is non-empty and not $\mathbb{C}$.

Theorem 3.1 (Riemann mapping theorem)

Suppose Ω is proper and simply connected. If $z_0 \in \Omega$, then there exists a unique conformal map $F : \Omega \to \mathbb{D}$ such that $F(z_0) = 0$ and $F'(z_0) > 0$.

Corollary 3.2

Any two proper simply connected open subsets in $\mathbb{C}$ are conformally equivalent.

The corollary follows via $\mathbb{D}$. *Uniqueness*: $H = F \circ G^{-1} \in \mathrm{Aut}(\mathbb{D})$ fixes 0, so $H(z) = e^{i\theta}z$, and $H'(0) > 0$ gives $F = G$. For existence, among injective holomorphic $f : \Omega \to \mathbb{D}$ with $f(z_0) = 0$ take one maximizing $|f'(z_0)|$; it fills $\mathbb{D}$. We need limits of sequences of functions.

3.2 Montel's theorem

A family $\mathcal{F}$ of holomorphic functions on Ω is **normal** if every sequence in $\mathcal{F}$ has a subsequence converging uniformly on compact subsets (limit not necessarily in $\mathcal{F}$); **uniformly bounded on compact subsets** if for each compact K some B bounds all $|f|$, $f \in \mathcal{F}$, on K; **equicontinuous** on compact K if for every $\epsilon > 0$ some $\delta > 0$ gives $|f(z) - f(w)| < \epsilon$ for all $f \in \mathcal{F}$ and $z, w \in K$, $|z-w| < \delta$. (Differentiable functions on $[0,1]$ with uniformly bounded derivatives are equicontinuous; x^n on $[0,1]$ are not.)

Theorem 3.3 (Montel)

Suppose $\mathcal{F}$ is a family of holomorphic functions on Ω that is uniformly bounded on compact subsets of Ω. Then:
(i) $\mathcal{F}$ is equicontinuous on every compact subset of Ω.
(ii) $\mathcal{F}$ is a normal family.

Part (i) uses the Cauchy integral formula (contrast $\sin(nx)$ on $(0,1)$: bounded, not equicontinuous); part (ii) is Arzelà–Ascoli. A sequence of compact $K_\ell \subset \Omega$ is an **exhaustion** if (a) K_ℓ lies in the interior of $K_{\ell+1}$, and (b) every compact $K \subset \Omega$ lies in some K_ℓ; so $\Omega = \bigcup_\ell K_\ell$.

Lemma 3.4

Any open set Ω in the complex plane has an exhaustion.

Proof. If Ω is bounded, let K_ℓ be the set of points of Ω at distance $\geq 1/\ell$ from $\partial\Omega$. If Ω is unbounded, require also $|z| \leq \ell$ for $z \in K_\ell$. ∎

Proof of Montel's theorem. (i) Let $K \subset \Omega$ be compact and $r > 0$ with $3r < d(K, \partial\Omega)$, so $D_{3r}(z) \subset \Omega$ for $z \in K$. Let $z, w \in K$ with $|z - w| < r$ and γ the boundary circle of $D_{2r}(w)$. By Cauchy's integral formula

$$f(z) - f(w) = \frac{1}{2\pi i}\int_\gamma f(\zeta)\Big[\frac{1}{\zeta - z} - \frac{1}{\zeta - w}\Big]d\zeta, \qquad \Big|\frac{1}{\zeta - z} - \frac{1}{\zeta - w}\Big| = \frac{|z-w|}{|\zeta - z||\zeta - w|} \leq \frac{|z-w|}{r^2},$$

since $\zeta \in \gamma$ and $|z - w| < r$. Hence $|f(z) - f(w)| \leq \frac{1}{2\pi} 2\pi(2r)\frac{B|z-w|}{r^2}$, B the uniform bound for $\mathcal{F}$ on the compact set of points at distance $\leq 2r$ from K. So $|f(z) - f(w)| < C|z - w|$ for all $f \in \mathcal{F}$: equicontinuity.

(ii) Let $\{f_n\} \subset \mathcal{F}$, $K \subset \Omega$ compact, and $\{w_j\}$ dense in Ω. By boundedness, successive subsequences converge at w_1, w_2, …; the diagonal $g_n = f_{n,n}$ converges at every w_j. Uniformly on K: given ϵ, take δ from equicontinuity, cover K by $D_\delta(w_1), \ldots, D_\delta(w_J)$, and choose N with $|g_m(w_j) - g_n(w_j)| < \epsilon$ for $n, m > N$, $j \leq J$. For $z \in K \cap D_\delta(w_j)$,

$$|g_n(z) - g_m(z)| \leq |g_n(z) - g_n(w_j)| + |g_n(w_j) - g_m(w_j)| + |g_m(w_j) - g_m(z)| < 3\epsilon.$$

Finally, with an exhaustion, diagonalize again: $\{g_{n,n}\}$ converges uniformly on every K_ℓ, hence on every compact subset. ∎

Proposition 3.5

If Ω is connected open and injective holomorphic f_n converge uniformly on compact subsets to f, then f is injective or constant.

Proof. Suppose $f(z_1) = f(z_2)$, $z_1 \neq z_2$. $g_n = f_n - f_n(z_1)$ vanishes only at z_1, and $g_n \to g = f - f(z_1)$. If $g \not\equiv 0$, z_2 is an isolated zero of g, so

$$1 \leq \frac{1}{2\pi i} \int_\gamma \frac{g'(\zeta)}{g(\zeta)}\, d\zeta,$$

for a small circle γ about z_2 with no other zeros of g on or inside it. As $1/g_n \to 1/g$, $g_n' \to g'$ uniformly on γ (Chapter 2, Theorem 5.3), $\frac{1}{2\pi i}\int_\gamma \frac{g_n'}{g_n} \to \frac{1}{2\pi i}\int_\gamma \frac{g'}{g}$; but the left side is 0 ($g_n \neq 0$ inside γ), a contradiction. So $g \equiv 0$ and f is constant. ■

3.3 Proof of the Riemann mapping theorem

Step 1. Ω is conformally equivalent to an open subset of $\mathbb{D}$ containing 0. Take $\alpha \notin \Omega$; on the simply connected Ω there is $f(z) = \log(z - \alpha)$, $e^f = z - \alpha$, so f is injective. Fix $w \in \Omega$: $f(z) \neq f(w) + 2\pi i$ (exponentiate), and $f(\Omega)$ stays away from $f(w) + 2\pi i$, since $f(z_n) \to f(w) + 2\pi i$ would give $z_n \to w$, so $f(z_n) \to f(w)$. Thus

$$F(z) = \frac{1}{f(z) - (f(w) + 2\pi i)}$$

is injective and bounded on Ω; translating and rescaling $F(\Omega)$ gives the claim.

Step 2. So assume $\Omega \subset \mathbb{D}$ is open with $0 \in \Omega$, and let

$$\mathcal{F} = \{f : \Omega \to \mathbb{D} \text{ holomorphic, injective, and } f(0) = 0\}.$$

$\mathcal{F} \neq \emptyset$ (identity) is uniformly bounded, and so are the $|f'(0)|$ (Cauchy inequalities). Let $s = \sup_{\mathcal{F}} |f'(0)| \geq 1$ and $|f_n'(0)| \to s$. By Montel a subsequence converges uniformly on compacta to f, non-constant as $s \geq 1$, hence injective (Proposition 3.5). $|f| \leq 1$ by continuity, $|f| < 1$ by maximum modulus, $f(0) = 0$: $f \in \mathcal{F}$ with $|f'(0)| = s$.

Step 3. f is surjective. Otherwise take $\alpha \in \mathbb{D} - f(\Omega)$ and $\psi_\alpha(z) = \frac{\alpha - z}{1 - \overline{\alpha} z}$. $U = (\psi_\alpha \circ f)(\Omega)$ is simply connected and omits 0, so $g(w) = e^{\frac{1}{2}\log w}$ is defined on U. Let

$$F = \psi_{g(\alpha)} \circ g \circ \psi_\alpha \circ f.$$

F is holomorphic, maps 0 to 0 ($g(\psi_\alpha(0)) = g(\alpha)$), maps Ω into $\mathbb{D}$, and is injective (each factor is). So $F \in \mathcal{F}$. With $h(w) = w^2$,

$$f = \psi_\alpha^{-1} \circ h \circ g^{-1} \circ \psi_{g(\alpha)}^{-1} \circ F = \Phi \circ F.$$

$\Phi = \psi_\alpha \circ h \circ \psi_{g(\alpha)}$ maps $\mathbb{D}$ into $\mathbb{D}$, $\Phi(0) = 0$, and is not injective (h is not), so $|\Phi'(0)| < 1$ by the Schwarz lemma. As $f'(0) = \Phi'(0)F'(0)$, $|f'(0)| < |F'(0)|$, contradicting maximality.

Finally multiply f by a constant of absolute value 1 so that $f'(0) > 0$. ■

Remark

Simple connectivity was used only for the logarithm and square root; *holomorphic simple connectivity* ($\int_\gamma f = 0$ for all holomorphic f and closed $\gamma \subset \Omega$) suffices. See Appendix B.

4 Conformal mappings onto polygons

For polygons the map is explicit: the *Schwarz–Christoffel formula*.

4.1 Some examples

Example 1

For $0 < \alpha < 2$, the map $f(z) = z^\alpha$ of $\mathbb{H}$ onto the sector $\{0 < \arg z < \alpha\pi\}$ (Section 1) can be written

$$z^\alpha = f(z) = \int_0^z f'(\zeta)\,d\zeta = \alpha \int_0^z \zeta^{-\beta}\,d\zeta, \qquad \alpha + \beta = 1,$$

integrated along any path in the closed upper half-plane. $\zeta^{-\beta}$ is integrable at 0 ($\beta < 1$), so $f(0) = 0$. For $x > 0$, $f'(x) = \alpha x^{\alpha-1} > 0$, not integrable at ∞, so $[0,\infty) \to [0,\infty)$. For $x < 0$, $f'(x) = -\alpha|x|^{\alpha-1}e^{i\pi\alpha}$, so $(-\infty, 0] \to (\infty e^{i\pi\alpha}, 0]$.

Example 2

For $z \in \mathbb{H}$ let $f(z) = \int_0^z \frac{d\zeta}{(1-\zeta^2)^{1/2}}$, the branch holomorphic in $\mathbb{H}$ and positive on $(-1,1)$; then $(1-\zeta^2)^{-1/2} = -i(\zeta^2-1)^{-1/2}$ for $\zeta > 1$ (similarly for $\zeta < -1$). $f(\pm 1) = \pm\pi/2$ and $f' > 0$ on $(-1,1)$, so $[-1,1] \to [-\pi/2, \pi/2]$; for $x > 1$, $f(x) = \frac{\pi}{2} + i\int_1^x \frac{dx}{(x^2-1)^{1/2}}$, which diverges, so $[1,\infty)$ goes onto the vertical half-line above $\pi/2$; similarly for $(-\infty, -1]$. In fact f inverts $\sin z$ (Example 8) and maps $\mathbb{H}$ conformally onto the half-strip.

Example 3

For fixed $0 < k < 1$, let

$$f(z) = \int_0^z \frac{d\zeta}{[(1-\zeta^2)(1-k^2\zeta^2)]^{1/2}}, \qquad z \in \mathbb{H},$$

the branch positive for $-1 < \zeta < 1$. Such integrals are **elliptic integrals** (variants arise in the arc length of an ellipse). With

$$K = \int_0^1 \frac{dx}{[(1-x^2)(1-k^2x^2)]^{1/2}}, \qquad K' = \int_1^{1/k} \frac{dx}{[(x^2-1)(1-k^2x^2)]^{1/2}},$$

f maps $\mathbb{R}$ onto the rectangle with vertices $-K$, K, $K+iK'$, $-K+iK'$. $f(\pm 1) = \pm K$, $f' > 0$ on $(-1,1)$: $[-1,1] \to [-K, K]$. For $1 < x < 1/k$, $f(x) = K + i\int_1^x \frac{d\xi}{[(\xi^2-1)(1-k^2\xi^2)]^{1/2}}$: $[1, 1/k] \to [K, K+iK']$; similarly $[-1/k, -1] \to [-K+iK', -K]$. For $x > 1/k$, $f(x) = K + iK' - \int_{1/k}^x \frac{dx}{[(x^2-1)(k^2x^2-1)]^{1/2}}$, and $\int_{1/k}^\infty \frac{dx}{[(x^2-1)(k^2x^2-1)]^{1/2}} = K$ ($x = 1/ku$), so $[1/k, \infty) \to [K+iK', iK']$; similarly $(-\infty, -1/k] \to [-K+iK', iK']$. ∞ goes to iK'.

Two problems: generalize (the Schwarz–Christoffel integral maps $\mathbb{R}$ to a polygonal line), and prove conformality from $\mathbb{H}$ onto the polygon's interior.

4.2 The Schwarz–Christoffel integral

Definition (Schwarz–Christoffel integral)

For real $A_1 < A_2 < \cdots < A_n$ and exponents $\beta_k < 1$ with $1 < \sum_{k=1}^n \beta_k$, let

$$S(z) = \int_0^z \frac{d\zeta}{(\zeta - A_1)^{\beta_1} \cdots (\zeta - A_n)^{\beta_n}}, \tag{5}$$

where $(z - A_k)^{\beta_k}$ is the branch in the plane slit along $\{A_k + iy : y \le 0\}$ that is positive for real $x > A_k$; thus $(x - A_k)^{\beta_k} = |x - A_k|^{\beta_k} e^{i\pi\beta_k}$ for real $x < A_k$.

The plane slit along the rays $\{A_k + iy : y \le 0\}$ is simply connected, so S is holomorphic there; the singularities are integrable ($\beta_k < 1$), so S is continuous up to $\mathbb{R}$, along any path avoiding the open slits. As $\prod_k |\zeta - A_k|^{-\beta_k} \le c|\zeta|^{-\sum \beta_k}$ with $\sum \beta_k > 1$, the integral converges at infinity and by Cauchy $a_\infty = \lim_{r\to\infty} S(re^{i\theta})$ exists independently of $0 \le \theta \le \pi$; let $a_k = S(A_k)$. (The cases $\sum \beta_k \le 1$ of Examples 1–2 need a modification, S then being unbounded.)

Proposition 4.1

Suppose $S(z)$ is given by (5).
(i) If $\sum_{k=1}^n \beta_k = 2$ and p is the polygon with vertices $a_1, \dots, a_n$ (in order), S maps $\mathbb{R}$ onto $p - \{a_\infty\}$, $a_\infty \in [a_n, a_1]$ being the image of ∞, and the interior angle at a_k is $\alpha_k\pi$, $\alpha_k = 1 - \beta_k$.
(ii) If $1 < \sum_{k=1}^n \beta_k < 2$, the image of the extended line is the $(n+1)$-gon $a_1, \dots, a_n, a_\infty$, with angle $\alpha_\infty\pi$ at a_∞, $\alpha_\infty = 1 - \beta_\infty$, $\beta_\infty = 2 - \sum_{k=1}^n \beta_k$.

Proof. Let $\sum \beta_k = 2$. For $A_k < x < A_{k+1}$,

$$S'(x) = \prod_{j\le k} (x - A_j)^{-\beta_j} \prod_{j>k} (x - A_j)^{-\beta_j}, \qquad \arg S'(x) = \arg \prod_{j>k} e^{-i\pi\beta_j} = -\pi \sum_{j>k} \beta_j,$$

constant on (A_k, A_{k+1}). As $S(x) = S(A_k) + \int_{A_k}^x S'(y)\,dy$, S moves along the segment $[a_k, a_{k+1}]$ ($[z, w] = \{(1-t)z + tw : t \in [0,1]\}$) at angle $-\pi \sum_{j>k} \beta_j$. For $x > A_n$, $S'(x) > 0$; for $x < A_1$, $\arg S'(x) = -2\pi$, so again $S' > 0$. So $[A_n, \infty)$ and $(-\infty, A_1]$ give $[a_n, a_\infty) \cup (a_\infty, a_1] = [a_n, a_1] - \{a_\infty\}$. The direction of $[a_k, a_{k+1}]$ exceeds that of $[a_{k-1}, a_k]$ by $\pi\beta_k$, so the interior angle at a_k is $\pi - \pi\beta_k = \alpha_k\pi$. The case $1 < \sum \beta_k < 2$ is similar. ∎

This does not settle the problem:

(1) for general n and A_k the image polygon need not be simple, nor S conformal on $\mathbb{H}$;

(2) it does not show that for a given polygonal region P, S (suitably modified) maps $\mathbb{H}$ conformally onto P. We now prove it does.

4.3 Boundary behavior

A **polygonal region** P is a bounded simply connected open set whose boundary is a closed polygonal line p (a *polygon*). For a conformal $f : U \to f(U)$,

$$\text{Area}(f(U)) = \iint_U |f'(z)|^2\,dx\,dy,$$

since the Jacobian of $w = f(z)$ is $|f'(z)|^2$ (Proposition 2.3 of Chapter 1).

Theorem 4.2

A conformal map $F : \mathbb{D} \to P$ extends to a continuous bijection $\overline{\mathbb{D}} \to \overline{P}$; in particular $\partial\mathbb{D}$ maps bijectively onto the polygon p.

Lemma 4.3

Let C_r $(0 < r < 1/2)$ be the circle of radius r about $z_0 \in \partial\mathbb{D}$, and $z_r, z_r' \in \mathbb{D} \cap C_r$ for all small r. With $\rho(r) = |F(z_r) - F(z_r')|$, there are $r_n \to 0$ with $\rho(r_n) \to 0$.

Proof. If not, there are $c > 0$, $0 < R < 1/2$ with $c \le \rho(r)$ for $0 < r \le R$. Now $F(z_r) - F(z_r') = \int_\gamma F'(\zeta)\,d\zeta$, γ the arc of C_r in $\mathbb{D}$ joining z_r, z_r'; parametrizing it by $z_0 + re^{i\theta}$, $\theta_1(r) \le \theta \le \theta_2(r)$,

$$\rho(r) \le \int_{\theta_1(r)}^{\theta_2(r)} |F'(z)| r\,d\theta \le \Big(\int_{\theta_1(r)}^{\theta_2(r)} |F'(z)|^2 r\,d\theta\Big)^{1/2} \Big(\int_{\theta_1(r)}^{\theta_2(r)} r\,d\theta\Big)^{1/2}$$

by Cauchy–Schwarz. Squaring and dividing by r, $\frac{\rho(r)^2}{r} \le 2\pi \int_{\theta_1(r)}^{\theta_2(r)} |F'(z)|^2 r\,d\theta$. Integrating from 0 to R,

$$c^2 \int_0^R \frac{dr}{r} \le 2\pi \int_0^R \int_{\theta_1(r)}^{\theta_2(r)} |F'(z)|^2 r\,d\theta\,dr \le 2\pi \iint_{\mathbb{D}} |F'(z)|^2\,dx\,dy = 2\pi\,\mathrm{Area}(P).$$

The left side is infinite and the right side finite—a contradiction. ∎

Lemma 4.4

Let z_0 be a point on the unit circle. Then $F(z)$ tends to a limit as z approaches z_0 within the unit disc.

Proof. Otherwise $z_n, z_n' \to z_0$ in $\mathbb{D}$ with $F(z_n) \to \zeta$, $F(z_n') \to \zeta'$, $\zeta \ne \zeta'$, both on p (conformality). Take disjoint discs D, D' about ζ, ζ' at distance $d > 0$, and continuous curves $\Lambda \subset D \cap P$, $\Lambda' \subset D' \cap P$ through $F(z_n)$, resp. $F(z_n')$, for all large n, ending at ζ, ζ'. Then $\lambda = F^{-1}(\Lambda)$, $\lambda' = F^{-1}(\Lambda')$ contain infinitely many z_n, resp. z_n', so for small r, C_r meets them at z_r, z_r' with $|F(z_r) - F(z_r')| > d$, contradicting Lemma 4.3. ∎

Lemma 4.5

The conformal map F extends to a continuous function from the closure of the disc to the closure of the polygon.

Proof. Define $F(z_0) = \lim_{z \to z_0} F(z)$ for $|z_0| = 1$ (Lemma 4.4). Given ϵ, there is δ with $|F(z) - F(z_0)| < \epsilon$ for $z \in \mathbb{D}$, $|z - z_0| < \delta$. If $|z| = 1$ and $|z - z_0| < \delta$, choose $w \in \mathbb{D}$ with $|F(z) - F(w)| < \epsilon$ and $|w - z_0| < \delta$; then $|F(z) - F(z_0)| \le |F(z) - F(w)| + |F(w) - F(z_0)| < 2\epsilon$. ∎

Proof of Theorem 4.2. So F extends continuously to $\overline{\mathbb{D}} \to \overline{P}$. The same argument applies to $G = F^{-1}$, since the property used ($C \cap \mathbb{D}$ is an arc for small circles C about boundary points) holds at every boundary point of P. If $z_k \to z \in \overline{\mathbb{D}}$, $G(F(z_k)) = z_k$ gives $G(F(z)) = z$; similarly $F(G(w)) = w$. ∎

4.4 The mapping formula

Let P be bounded by p with consecutive vertices $a_1, \dots, a_n$, $n \ge 3$, interior angles $\alpha_k\pi$, exterior angles $\beta_k\pi$, $\alpha_k + \beta_k = 1$; then $\sum \beta_k = 2$. Via $w = (i - z)/(i + z)$ $(\infty \mapsto -1)$, a conformal map $F : \mathbb{H} \to P$ (Riemann) extends to a continuous bijection of $\mathbb{R} \cup \{\infty\}$ onto p. First assume no

vertex corresponds to ∞: $F(A_k) = a_k$; after relabeling $A_1 < \cdots < A_n$, and F maps $[A_k, A_{k+1}]$ onto $[a_k, a_{k+1}]$ and $(-\infty, A_1] \cup [A_n, \infty)$ onto $[a_n, a_1]$.

Theorem 4.6 (Schwarz–Christoffel)

There are $c_1, c_2 \in \mathbb{C}$ with $F(z) = c_1 S(z) + c_2$, S the Schwarz–Christoffel integral of Section 4.2.

Proof. Let $1 < k < n$. F maps $[A_{k-1}, A_k]$ and $[A_k, A_{k+1}]$ onto two segments meeting at $a_k = F(A_k)$ at angle $\alpha_k \pi$. Choosing a branch of the logarithm, define

$$h_k(z) = (F(z) - a_k)^{1/\alpha_k}$$

on the half-strip of $\mathbb{H}$ over (A_{k-1}, A_{k+1}). h_k is continuous up to (A_{k-1}, A_{k+1}), mapping it into a straight segment L_k with $A_k \mapsto 0$, so by reflection it continues to the strip $A_{k-1} < \operatorname{Re}(z) < A_{k+1}$. $h_k' \neq 0$ there: in the upper half-strip,

$$\frac{F'(z)}{F(z) - F(A_k)} = \frac{1}{\alpha_k} \frac{h_k'(z)}{h_k(z)},$$

and $F'(z) \neq 0$ (Proposition 1.1), so $h_k' \neq 0$, and by reflection in the lower half-strip too. For $A_{k-1} < x < A_{k+1}$, h_k maps a small half-disc about x to one side of L_k injectively, hence (reflection) is injective on the whole disc, so $h_k'(x) \neq 0$.

Since $F = a_k + h_k^{\alpha_k}$, $F' = \alpha_k h_k^{\alpha_k - 1} h_k'$ and $F'' = \alpha_k(\alpha_k - 1) h_k^{\alpha_k - 2}(h_k')^2 + \alpha_k h_k^{\alpha_k - 1} h_k''$; as $h_k' \neq 0$ and h_k has a simple zero at A_k,

$$\frac{F''(z)}{F'(z)} = \frac{-\beta_k}{z - A_k} + E_k(z),$$

E_k holomorphic in the strip $A_{k-1} < \operatorname{Re}(z) < A_{k+1}$. Similarly $\frac{F''}{F'} = \frac{-\beta_1}{z - A_1} + E_1$ for $\operatorname{Re}(z) < A_2$ and $\frac{F''}{F'} = \frac{-\beta_n}{z - A_n} + E_n$ for $\operatorname{Re}(z) > A_{n-1}$. F also reflects across $(-\infty, A_1) \cup (A_n, \infty)$, so it continues boundedly to $|z| > R$, hence is holomorphic at infinity: $F(z) = c_0 + \frac{c_1}{z} + \cdots$, and $F''/F' = O(1/z)$. As the strips cover $\mathbb{C}$,

$$\frac{F''(z)}{F'(z)} + \sum_{k=1}^{n} \frac{\beta_k}{z - A_k}$$

is entire and vanishes at infinity; by Liouville it is zero. With $Q(z) = (z - A_1)^{-\beta_1} \cdots (z - A_n)^{-\beta_n}$, $\frac{Q'}{Q} = -\sum_k \frac{\beta_k}{z - A_k} = \frac{F''}{F'}$, so $\frac{d}{dz}\left(\frac{F'(z)}{Q(z)}\right) = 0$ and $F' = cQ$. A final integration yields the theorem. ∎

Theorem 4.7

If a conformal $F : \mathbb{H} \to P$ maps $A_1, \ldots, A_{n-1}, \infty$ to the vertices of p, there are constants C_1, C_2 with

$$F(z) = C_1 \int_0^z \frac{d\zeta}{(\zeta - A_1)^{\beta_1} \cdots (\zeta - A_{n-1})^{\beta_{n-1}}} + C_2.$$

So the formula is obtained by deleting the last factor of the Schwarz–Christoffel integral.

Proof. Translate so that $A_j \neq 0$. Choose a real $A_n^* > 0$ (different from the others; [Supplement]), $\Phi(z) = A_n^* - \frac{1}{z} \in \operatorname{Aut}(\mathbb{H})$, $A_k^* = \Phi(A_k)$, $A_n^* = \Phi(\infty)$. Then $(F \circ \Phi^{-1})(A_k^*) = a_k$, $k = 1, \ldots, n$, and by Theorem 4.6

$$(F \circ \Phi^{-1})(z') = C_1' \int_0^{z'} \frac{d\zeta}{(\zeta - A_1^*)^{\beta_1} \cdots (\zeta - A_n^*)^{\beta_n}} + C_2'.$$

Change variables $\zeta = \Phi(w)$, $d\zeta = dw/w^2$; since $\zeta - A_n^* = -1/w$ and $\zeta - A_k^* = (w(A_n^* - A_k^*) - 1)/w$, and $2 = \beta_1 + \cdots + \beta_n$, the integrand becomes $C \prod_{k<n} (w - 1/(A_n^* - A_k^*))^{-\beta_k}$ up to a constant factor. Since $1/(A_n^* - A_k^*) = A_k$, setting $\Phi^{-1}(z') = z$ gives the formula. ∎

4.5 Return to elliptic integrals

$I(z) = \int_0^z \frac{d\zeta}{[(1-\zeta^2)(1-k^2\zeta^2)]^{1/2}}$, $0 < k < 1$, maps $\mathbb{R}$ onto the rectangle R with vertices $-K, K, K + iK', -K + iK'$ (Example 3); we show it maps $\mathbb{H}$ conformally onto R. By Theorem 4.6 there is a conformal $F : \mathbb{H} \to R$ sending four real points to the vertices, and after an automorphism $-1, 0, 1 \mapsto -K, 0, K$. Let $F(1/\ell) = K + iK'$, $0 < \ell < 1$. $\tilde{F}(z) = -\overline{F(-\overline{z})}$ is also such a map (symmetry of R), so $F^{-1} \circ \tilde{F}$ fixes $-1, 0, 1$, is the identity, and $F(-1/\ell) = -K + iK'$. By Theorem 4.6,

$$F(z) = c_1 \int_0^z \frac{d\zeta}{[(1-\zeta^2)(1-\ell^2\zeta^2)]^{1/2}} + c_2.$$

$z = 0$ gives $c_2 = 0$; $z = 1$, $1/\ell$ give $K(k) = c_1 K(\ell)$, $K'(k) = c_1 K'(\ell)$, with

$$K(k) = \int_0^1 \frac{dx}{[(1-x^2)(1-k^2x^2)]^{1/2}}, \qquad K'(k) = \int_1^{1/k} \frac{dx}{[(x^2-1)(1-k^2x^2)]^{1/2}}.$$

K is strictly increasing and $K'(k) = K(\tilde{k})$, $\tilde{k}^2 = 1 - k^2$, is strictly decreasing, so K/K' is strictly increasing; $K(k)/K'(k) = K(\ell)/K'(\ell)$ forces $k = \ell$, $c_1 = 1$, $I = F$.

Let sn be the inverse of I (Jacobi's notation, cf. sin); it maps $R_0 = R$ onto the closed upper half-plane. Reflecting across the lower sides gives rectangles $R_1, R_2, \ldots$ with $\mathrm{sn}(z) = \overline{\mathrm{sn}(\overline{z})}$ on R_1, $\mathrm{sn}(z) = \mathrm{sn}(-iK' + \overline{z})$ on R_2, and so on, extending sn to $-K < \mathrm{Re}(z) < K$ with period $2iK'$. Reflecting in horizontal rows extends it to $\mathbb{C}$ with period $4K$. So sn is *doubly periodic* (periods $4K$, $2iK'$; meromorphic, two simple poles per period parallelogram): Chapter 9.

Chapter 9. An Introduction to Elliptic Functions

Overview

Elliptic functions grew out of *elliptic integrals* $\int R(x, \sqrt{P(x)})\,dx$, R rational, P of degree three or four (arc length of an ellipse or lemniscate), and their double periodicity (cf. Section 4). Jacobi built the theory on theta functions; we sketch Weierstrass's simpler approach via $\wp$, reaching the *Eisenstein series* and divisor functions. Theta functions follow in Chapter 10.

1 Elliptic functions

Let f be meromorphic on $\mathbb{C}$ and **doubly periodic**: $f(z+\omega_1) = f(z+\omega_2) = f(z)$ for non-zero ω_1, ω_2. If $\omega_2/\omega_1 \in \mathbb{R}$, f is simply periodic or constant, so assume ω_1, ω_2 are $\mathbb{R}$-independent. *Normalization*: with $\tau = \omega_2/\omega_1$, interchanging if needed (τ, $1/\tau$ have imaginary parts of opposite signs), $\mathrm{Im}(\tau) > 0$; f has periods ω_1, ω_2 iff $F(z) = f(\omega_1 z)$ has periods $1, \tau$. So assume periods 1 and τ, $\mathrm{Im}(\tau) > 0$. Then

$$f(z + n + m\tau) = f(z) \qquad \text{for all integers } n, m \text{ and all } z \in \mathbb{C}, \tag{1}$$

and we consider the **lattice** $\Lambda = \{n+m\tau : n, m \in \mathbb{Z}\}$ **generated** by 1 and τ, and the **fundamental parallelogram**

$$P_0 = \{z \in \mathbb{C} : z = a + b\tau,\ 0 \le a < 1,\ 0 \le b < 1\}.$$

$z \sim w$ (**congruent modulo** Λ) if $z - w \in \Lambda$; then $f(z) = f(w)$ by (1). Every z is congruent to a unique point of P_0: with $z = a+b\tau$ ($a, b \in \mathbb{R}$) and n, m the integer parts of a, b, $w = z-n-m\tau \in P_0$; if $a + b\tau \sim a' + b'\tau$ in P_0, then $a - a', b - b' \in \mathbb{Z} \cap (-1, 1) = \{0\}$. A **period parallelogram** is a translate $P = P_0 + h$; applied to $z - h$, the same holds for P. Moreover

$$\mathbb{C} = \bigcup_{n,m \in \mathbb{Z}} (n + m\tau + P_0), \tag{2}$$

a disjoint union.

Proposition 1.1

Suppose f is a meromorphic function with two periods 1 and τ which generate the lattice Λ. Then:
(i) Every point in $\mathbb{C}$ is congruent to a unique point in the fundamental parallelogram.
(ii) Every point in $\mathbb{C}$ is congruent to a unique point in any given period parallelogram.
(iii) The lattice Λ provides a disjoint covering of the complex plane, in the sense of (2).
(iv) The function f is completely determined by its values in any period parallelogram.

1.1 Liouville's theorems

Theorem 1.2

An entire doubly periodic function is constant.

Proof. It is determined by its values on P_0, whose closure is compact; so it is bounded on $\mathbb{C}$, hence constant by Liouville's theorem. ∎

A non-constant doubly periodic meromorphic function is an **elliptic function**; it has finitely many zeros and poles in a period parallelogram, counted with multiplicity.

Theorem 1.3

The total number of poles of an elliptic function in P_0 is always ≥ 2.

So f cannot have just one simple pole (a single double pole is allowed).

Proof. Suppose first f has no poles on ∂P_0. By the residue theorem $\int_{\partial P_0} f(z)\,dz = 2\pi i \sum \operatorname{res} f$, and the integral is 0 by periodicity:

$$\int_{\partial P_0} f = \int_0^1 f + \int_1^{1+\tau} f + \int_{1+\tau}^{\tau} f + \int_{\tau}^0 f,$$

and opposite sides cancel; e.g.

$$\int_0^1 f(z)\,dz + \int_{1+\tau}^{\tau} f(z)\,dz = \int_0^1 f(z)\,dz + \int_1^0 f(s+\tau)\,ds = \int_0^1 f(z)\,dz - \int_0^1 f(s)\,ds = 0.$$

So $\sum \operatorname{res} f = 0$, and f has at least two poles in P_0. With a pole on ∂P_0, use $P = h + P_0$ for a small h. ■

The total number of poles (with multiplicities) of an elliptic function is its **order**.

Theorem 1.4

Every elliptic function of order m has m zeros in P_0.

Proof. With no zeros or poles on ∂P_0, $\int_{\partial P_0} \frac{f'}{f} = 2\pi i(N_z - N_p)$ vanishes by periodicity; otherwise use a translate. ■

Consequently, for every $c \in \mathbb{C}$, $f(z) = c$ has as many solutions in P_0 as the order of f ($f - c$ is elliptic with the same poles).

1.2 The Weierstrass $\wp$ function

An elliptic function of order two. We want double poles exactly at the lattice points. For one period, $\sum_n \frac{1}{z+n}$ is 1-periodic with poles at the integers but not absolutely convergent; summing symmetrically,

$$F(z) = \lim_{N\to\infty} \sum_{|n|\leq N} \frac{1}{z+n} = \frac{1}{z} + \sum_{n=1}^{\infty} \Big[\frac{1}{z+n} + \frac{1}{z-n}\Big],$$

the bracket being $O(1/n^2)$; $F(z) = \pi \cot \pi z$ (Chapter 5). Equivalently $\frac{1}{z} + \sum_{n\neq 0} \big[\frac{1}{z+n} - \frac{1}{n}\big]$, absolutely convergent, since pairing $\pm n$ gives $\frac{1}{z+n} + \frac{1}{z-n}$. Mimicking this, as $\sum_{\omega\in\Lambda} \frac{1}{(z+\omega)^2}$ is not absolutely convergent, with $\Lambda^* = \Lambda - \{0\}$ consider

$$\frac{1}{z^2} + \sum_{\omega\in\Lambda^*} \Big[\frac{1}{(z+\omega)^2} - \frac{1}{\omega^2}\Big], \qquad \frac{1}{(z+\omega)^2} - \frac{1}{\omega^2} = \frac{-z^2 - 2z\omega}{(z+\omega)^2\omega^2} = O\Big(\frac{1}{|\omega|^3}\Big) \text{ as } |\omega| \to \infty.$$

Lemma 1.5

The two series

$$\sum_{(n,m)\neq(0,0)} \frac{1}{(|n|+|m|)^r} \qquad \text{and} \qquad \sum_{n+m\tau\in\Lambda^*} \frac{1}{|n+m\tau|^r}$$

converge if $r > 2$.

(By the Note at the end of Chapter 7 the order of summation does not matter; we sum in m, then n.)

Proof. For $n \neq 0$, by integral comparison ($1/k^r \le \int_{k-1}^{k} dx/x^r$),

$$\sum_{m\in\mathbb{Z}} \frac{1}{(|n|+|m|)^r} = \frac{1}{|n|^r} + 2\sum_{k\ge|n|+1} \frac{1}{k^r} \le \frac{1}{|n|^r} + 2\int_{|n|}^{\infty} \frac{dx}{x^r} \le \frac{1}{|n|^r} + \frac{C}{|n|^{r-1}},$$

so for $r>2$, $\sum_{(n,m)\neq(0,0)} \frac{1}{(|n|+|m|)^r} = \sum_{|m|\neq 0} \frac{1}{|m|^r} + \sum_{|n|\neq 0}\sum_m \frac{1}{(|n|+|m|)^r} < \infty$. For the second series it suffices that $|n|+|m| \le c|n+m\tau|$. Write $A \approx B$ if $A \lesssim B \lesssim A$ ($x \lesssim y$: $x \le ay$ for some $a>0$). For $A, B > 0$, $(A^2+B^2)^{1/2} \approx A+B$ (square; and $A, B \le (A^2+B^2)^{1/2}$). With $\tau = s+it$, $t>0$,

$$|n+m\tau| = [(n+ms)^2 + (mt)^2]^{1/2} \approx |n+ms| + |mt| \approx |n+ms| + |m|,$$

and $|n+ms|+|m| \approx |n|+|m|$, considering separately $|n| \ge 2|m||s|$ and $|n| < 2|m||s|$. ∎

Remark

The proof shows that for $r>2$ the series $\sum |n+m\tau|^{-r}$ converges uniformly in every half-plane $\mathrm{Im}(\tau) \ge \delta > 0$. For $r=2$ it diverges.

The **Weierstrass $\wp$ function** is

$$\wp(z) = \frac{1}{z^2} + \sum_{\omega\in\Lambda^*} \Big[\frac{1}{(z+\omega)^2} - \frac{1}{\omega^2}\Big] = \frac{1}{z^2} + \sum_{(n,m)\neq(0,0)} \Big[\frac{1}{(z+n+m\tau)^2} - \frac{1}{(n+m\tau)^2}\Big].$$

It is meromorphic with double poles at the lattice points: for $|z| < R$ the terms with $|\omega| > 2R$ are $O(1/|\omega|^3)$ uniformly, so their sum is holomorphic (Lemma 1.5), and the finitely many others give the double poles.

Theorem 1.6

The function $\wp$ is an elliptic function that has periods 1 and τ, and double poles at the lattice points.

Proof. It remains to prove periodicity. Differentiating termwise,

$$\wp'(z) = -2\sum_{n,m\in\mathbb{Z}} \frac{1}{(z+n+m\tau)^3},$$

which converges absolutely for $z \notin \Lambda$ ($r=3$ in Lemma 1.5) and has periods 1, τ. So $\wp(z+1) = \wp(z)+a$, $\wp(z+\tau) = \wp(z)+b$. $\wp$ is even ($\omega \mapsto -\omega$), so $z=-1/2$, $z=-\tau/2$ give $a=b=0$. ∎

Properties of $\wp$. $\wp$ is even, so $\wp'$ is odd; with periodicity,

$$\wp'(1/2) = \wp'(\tau/2) = \wp'\Big(\frac{1+\tau}{2}\Big) = 0,$$

e.g. $\wp'(1/2) = -\wp'(-1/2) = -\wp'(-1/2+1) = -\wp'(1/2)$. Since $\wp'$ is elliptic of order 3, the **half-periods** $1/2$, $\tau/2$, $(1+\tau)/2$ are its only roots in P_0, each of multiplicity 1. Let

$$\wp(1/2) = e_1, \qquad \wp(\tau/2) = e_2, \qquad \wp\Big(\frac{1+\tau}{2}\Big) = e_3.$$

$\wp(z) = e_1$ has a double root at $1/2$ and, $\wp$ having order 2, no others in P_0; likewise e_2, e_3. So e_1, e_2, e_3 are distinct.

Theorem 1.7

The function $(\wp')^2$ is the cubic polynomial in $\wp$

$$(\wp')^2 = 4(\wp - e_1)(\wp - e_2)(\wp - e_3).$$

Proof. $F = (\wp - e_1)(\wp - e_2)(\wp - e_3)$ and $(\wp')^2$ both have, in P_0, only double roots at $1/2$, $\tau/2$, $(1+\tau)/2$ and poles of order 6 at the lattice points, so $(\wp')^2/F$ is constant; from $\wp = z^{-2} + \cdots$, $\wp' = -2z^{-3} + \cdots$ it is 4. ■

$\wp$ is universal:

Theorem 1.8

Every elliptic function f with periods 1 and τ is a rational function of $\wp$ and $\wp'$.

Lemma 1.9

Every even elliptic function F with periods 1 and τ is a rational function of $\wp$.

Proof. A zero or pole of the even F at 0 has even order, so replacing F by $\wp^m F$ we may assume none at the lattice points. Recall $\wp(z) - \wp(a)$ has a double zero if a is a half-period and zeros a, $-a$ otherwise. Zeros of F come in pairs $a, -a$ ($a \sim -a$ iff a is a half-period, then of even order). So if $a_1, -a_1, \ldots, a_m, -a_m$ are all zeros in a period parallelogram (with multiplicities, half for each of $a_j, -a_j$ at a half-period), $[\wp(z) - \wp(a_1)] \cdots [\wp(z) - \wp(a_m)]$ has exactly the roots of F. Similarly with the poles $b_1, -b_1, \ldots, b_m, -b_m$,

$$G(z) = \frac{[\wp(z) - \wp(a_1)] \cdots [\wp(z) - \wp(a_m)]}{[\wp(z) - \wp(b_1)] \cdots [\wp(z) - \wp(b_m)]}$$

is periodic with the same zeros and poles as F. So F/G is holomorphic and doubly periodic, hence constant. ■

Proof of Theorem 1.8. Write $f = f_{\text{even}} + f_{\text{odd}}$ with $f_{\text{even}}(z) = \frac{f(z)+f(-z)}{2}$, $f_{\text{odd}}(z) = \frac{f(z)-f(-z)}{2}$. Since $\wp'$ is odd, $f_{\text{odd}}/\wp'$ is even; apply the lemma to f_{even} and $f_{\text{odd}}/\wp'$. ■

2 The modular character of elliptic functions and Eisenstein series

Now write $\wp_\tau$. As $1, \tau$ and $1, \tau + 1$ generate the same lattice, $\wp_\tau = \wp_{\tau+1}$; $-1/\tau$ essentially interchanges the periods, and $\wp_{-1/\tau}(z) = \tau^2 \wp_\tau(\tau z)$. So we consider the **modular group** generated by $\tau \mapsto \tau + 1$ and $\tau \mapsto -1/\tau$ on $\mathbb{H}$.

2.1 Eisenstein series

The **Eisenstein series of order** k is

$$E_k(\tau) = \sum_{(n,m)\neq(0,0)} \frac{1}{(n + m\tau)^k} = \sum_{\omega \in \Lambda^*} \frac{1}{\omega^k},$$

for integers $k \geq 3$ and $\operatorname{Im}(\tau) > 0$.

Theorem 2.1

Eisenstein series have the following properties:
(i) The series $E_k(\tau)$ converges if $k \geq 3$, and is holomorphic in the upper half-plane.
(ii) $E_k(\tau) = 0$ if k is odd.
(iii) $E_k(\tau)$ satisfies the following transformation relations: $E_k(\tau+1) = E_k(\tau)$ and $E_k(\tau) = \tau^{-k}E_k(-1/\tau)$.

Property (iii) is the **modular character** of the Eisenstein series.

Proof. (i) By Lemma 1.5 and the remark after it, the series converges absolutely and uniformly in every half-plane $\mathrm{Im}(\tau) \geq \delta > 0$ for $k \geq 3$.

(ii) Replacing (n, m) by $(-n, -m)$ shows $E_k = -E_k$ for odd k.

(iii) $n + m(\tau+1) = (n+m) + m\tau$, and we may replace $n+m$ by n. Also $(n + m(-1/\tau))^k = \tau^{-k}(n\tau - m)^k$, and rearranging, replacing $(-m, n)$ by (n, m), gives the second relation. ∎

(Because of (ii), some authors define the Eisenstein series as $\sum 1/(n+m\tau)^{2k}$, possibly with a constant factor.)

Theorem 2.2

For z near 0, we have

$$\wp(z) = \frac{1}{z^2} + 3E_4 z^2 + 5E_6 z^4 + \cdots = \frac{1}{z^2} + \sum_{k=1}^{\infty}(2k+1)E_{2k+2}z^{2k}.$$

Proof. Replacing ω by $-\omega$, $\wp(z) = \frac{1}{z^2} + \sum_{\omega\in\Lambda^*}\left[\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2}\right]$. Differentiating the geometric series, $\frac{1}{(1-w)^2} = \sum_{\ell=0}^{\infty}(\ell+1)w^\ell$ for $|w| < 1$, so for small z

$$\frac{1}{(z-\omega)^2} = \frac{1}{\omega^2}\sum_{\ell=0}^{\infty}(\ell+1)\Big(\frac{z}{\omega}\Big)^\ell = \frac{1}{\omega^2} + \sum_{\ell=1}^{\infty}(\ell+1)\frac{z^\ell}{\omega^{\ell+2}}.$$

Hence

$$\wp(z) = \frac{1}{z^2} + \sum_{\ell=1}^{\infty}(\ell+1)\Big(\sum_{\omega\in\Lambda^*}\frac{1}{\omega^{\ell+2}}\Big)z^\ell = \frac{1}{z^2} + \sum_{\ell=1}^{\infty}(\ell+1)E_{\ell+2}z^\ell = \frac{1}{z^2} + \sum_{k=1}^{\infty}(2k+1)E_{2k+2}z^{2k},$$

since $E_{\ell+2} = 0$ for odd ℓ. ∎

Hence near 0,

$$\wp'(z) = \frac{-2}{z^3} + 6E_4 z + 20E_6 z^3 + \cdots,$$
$$(\wp'(z))^2 = \frac{4}{z^6} - \frac{24E_4}{z^2} - 80E_6 + \cdots, \qquad (\wp(z))^3 = \frac{1}{z^6} + \frac{9E_4}{z^2} + 15E_6 + \cdots.$$

So $(\wp')^2 - 4\wp^3 + 60E_4\wp + 140E_6$ is holomorphic near 0 and vanishes there; being doubly periodic, it is constant (Theorem 1.2), hence 0:

Corollary 2.3

If $g_2 = 60E_4$ and $g_3 = 140E_6$, then $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$.

This is another version of Theorem 1.7, and expresses the symmetric functions of the e_j via Eisenstein series.

2.2 Eisenstein series and divisor functions

Consider the Fourier expansion of the periodic function $E_k(\tau)$; equivalently, the Laurent expansion in $z = e^{2\pi i\tau}$.

Lemma 2.4

If $k \geq 2$ and $\mathrm{Im}(\tau) > 0$, then

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+\tau)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i \ell \tau}.$$

Proof. One may apply Poisson summation to $f(z) = 1/(z+\tau)^k$. Alternatively, it suffices to treat $k = 2$, the other cases following by termwise differentiation. Differentiating $\sum_{n=-\infty}^{\infty} \frac{1}{n+\tau} = \pi \cot \pi\tau$ (Chapter 5) gives

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+\tau)^2} = \frac{\pi^2}{\sin^2(\pi\tau)}.$$

With $w = e^{2\pi i\tau}$ ($|w| < 1$), $\sin^2 \pi\tau = -\frac{(e^{i\pi\tau} - e^{-i\pi\tau})^2}{4} = -\frac{(1-w)^2}{4w}$, so $\frac{\pi^2}{\sin^2 \pi\tau} = -4\pi^2 \frac{w}{(1-w)^2} = (-2\pi i)^2 \sum_{r=1}^{\infty} r w^r$, using $\sum_{r \geq 1} r w^r = \frac{w}{(1-w)^2}$. ∎

The **divisor function** $\sigma_\ell(r) = \sum_{d|r} d^\ell$ is the sum of the ℓth powers of the divisors of r.

Theorem 2.5

If $k \geq 4$ is even, and $\mathrm{Im}(\tau) > 0$, then

$$E_k(\tau) = 2\zeta(k) + \frac{2(-1)^{k/2}(2\pi)^k}{(k-1)!} \sum_{r=1}^{\infty} \sigma_{k-1}(r) e^{2\pi i \tau r}.$$

Proof. $\sigma_{k-1}(r) \leq \sum_{d \leq r} d^{k-1} \leq r^k$; if $\mathrm{Im}(\tau) \geq t_0 > 0$, $|e^{2\pi i r\tau}| \leq e^{-2\pi r t_0}$, so the series converges absolutely, by comparison with $\sum r^k e^{-2\pi r t_0}$. Using the definition, k even, and the lemma with τ replaced by $m\tau$,

$$\begin{aligned} E_k(\tau) &= \sum_{n \neq 0} \frac{1}{n^k} + \sum_{m \neq 0} \sum_{n=-\infty}^{\infty} \frac{1}{(n+m\tau)^k} = 2\zeta(k) + 2 \sum_{m>0} \sum_{n=-\infty}^{\infty} \frac{1}{(n+m\tau)^k} \\ &= 2\zeta(k) + 2\frac{(-2\pi i)^k}{(k-1)!} \sum_{m>0} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i \ell m \tau} = 2\zeta(k) + \frac{2(-1)^{k/2}(2\pi)^k}{(k-1)!} \sum_{r=1}^{\infty} \sigma_{k-1}(r) e^{2\pi i r \tau}, \end{aligned}$$

collecting the terms with $\ell m = r$. ∎

For the "forbidden" case $k = 2$, $\sum_{(n,m) \neq (0,0)} 1/(n+m\tau)^2$ no longer converges absolutely, but define

$$F(\tau) = \sum_m \sum_n \frac{1}{(n+m\tau)^2}, \qquad (n,m) \neq (0,0),$$

summed in the indicated order (first n, then m).

Corollary 2.6

The double sum defining F converges in the indicated order. We have

$$F(\tau) = 2\zeta(2) - 8\pi^2 \sum_{r=1}^{\infty} \sigma(r) e^{2\pi i r \tau},$$

where $\sigma(r) = \sum_{d|r} d$ is the sum of the divisors of r.

(Theorem 2.5's argument applies with $k = 2$.) $F(-1/\tau)\tau^{-2} \neq F(\tau)$: summing first in m then in n changes the value. This series is nevertheless crucial for the four-squares theorem (Chapter 10).

3 Selected exercises

Motivation. Exercises from Stein–Shakarchi (numbers kept).

Exercise 9.1 (Real ratio of periods)

Let f be meromorphic on $\mathbb{C}$ with non-zero periods ω_1, ω_2 such that $\omega_2/\omega_1 \in \mathbb{R}$.
(a) If $\omega_2/\omega_1 = p/q$ with p, q relatively prime integers, $q \geq 1$, show that f has the periods ω_1, ω_2 if and only if f has the single period $\omega_0 = \omega_1/q$.
(b) If ω_2/ω_1 is irrational, show that f is constant.

Solution. If ω, ω' are periods, so are $k\omega$, $k \in \mathbb{Z}$ (induction, and $f(z-\omega) = f((z-\omega)+\omega)$) and $\omega + \omega'$; so all integer combinations are periods.

(a)

Step 1. If ω_0 is a period, then $\omega_1 = q\omega_0$ and $\omega_2 = \frac{p}{q}\omega_1 = p\omega_0$ are integer multiples of it, hence periods.

Step 2. Conversely, since $\gcd(p, q) = 1$ there are integers m, n with $mq + np = 1$. Then

$$m\omega_1 + n\omega_2 = \omega_1\Big(m + n\frac{p}{q}\Big) = \frac{\omega_1}{q}(mq + np) = \omega_0,$$

an integer combination of ω_1, ω_2, hence a period.

(b) Let $\alpha = \omega_2/\omega_1 \notin \mathbb{Q}$ and $D = \{m + n\alpha : m, n \in \mathbb{Z}\}$.

Step 1. (D is dense in $\mathbb{R}$.) Of the $N+1$ fractional parts of $0, \alpha, \dots, N\alpha$, two (of $j\alpha$, $j'\alpha$) lie in one $[i/N, (i+1)/N)$, so $\delta = (j - j')\alpha - \ell$ has $0 < |\delta| < 1/N$ (α irrational). The multiples $k\delta \in D$ meet every interval of length $1/N$.

Step 2. (Many small periods.) For $t = m + n\alpha \in D$, $t\omega_1 = m\omega_1 + n\omega_2$ is a period. By Step 1 choose $t_k \in D \setminus \{0\}$ with $t_k \to 0$.

Step 3. (Conclusion.) The pole set S is discrete and closed, so $U = \mathbb{C} \setminus S$ is open and connected. For $z_0 \in U$, $z_k = z_0 + t_k\omega_1 \to z_0$, $z_k \neq z_0$, $f(z_k) = f(z_0)$ (Step 2), so $f \equiv f(z_0)$ on U; then f has no poles, $S = \varnothing$, and f is constant. ∎

Exercise 9.2 (Zeros minus poles)

Let f be elliptic with periods $1, \tau$, lattice Λ, zeros $a_1, \dots, a_r$ and poles $b_1, \dots, b_r$ in P_0 with multiplicity (equally many, Theorem 1.4). Show that

$$a_1 + \cdots + a_r - b_1 - \cdots - b_r \in \Lambda.$$

Solution.

Step 1. (Choice of a parallelogram.) For each of the countably many zeros and poles w, the h with $w \in \partial(h + P_0)$ form $w - \partial P_0$, of area zero; so some h puts none on ∂P, $P = h + P_0$.

Step 2. (Reduction to P.) By Proposition 1.1 there is a bijection $\phi : P_0 \to P$ with $\phi(z) - z \in \Lambda$, preserving orders of zeros and poles. So the zeros and poles in P are $a_i' = \phi(a_i)$, $b_j' = \phi(b_j)$, and

$$\textstyle\sum_i a_i - \sum_j b_j = \sum_i a_i' - \sum_j b_j' - \lambda \qquad \text{for some } \lambda \in \Lambda.$$

So it suffices to show $\sum_i a_i' - \sum_j b_j' \in \Lambda$.

Step 3. (Residues.) At a zero a of order k, zf'/f has residue ka; at a pole b of order k, $-kb$; elsewhere near $\overline{P}$ it is holomorphic. By the residue theorem,

$$\frac{1}{2\pi i}\int_{\partial P} z\frac{f'(z)}{f(z)}\,dz = \sum_i a_i' - \sum_j b_j'.$$

Step 4. (Pairing opposite sides.) Let $g = f'/f$, which has periods 1 and τ. The boundary runs $h \to h+1 \to h+1+\tau \to h+\tau \to h$. Substituting $z \mapsto z+\tau$ in the third side and $z \mapsto z+1$ in the second,

$$\int_h^{h+1} zg\,dz + \int_{h+1+\tau}^{h+\tau} zg\,dz = \int_h^{h+1} [z-(z+\tau)]g(z)\,dz = -\tau\int_h^{h+1} g(z)\,dz,$$
$$\int_{h+1}^{h+1+\tau} zg\,dz + \int_{h+\tau}^{h} zg\,dz = \int_h^{h+\tau} [(z+1)-z]g(z)\,dz = \int_h^{h+\tau} g(z)\,dz.$$

Step 5. (Integers.) The curve $\gamma(t) = f(h+t)$, $0 \le t \le 1$, is closed because $f(h+1) = f(h)$, and avoids 0. Hence

$$\int_h^{h+1} \frac{f'(z)}{f(z)}\,dz = \int_\gamma \frac{dw}{w} = 2\pi i\, n$$

with $n \in \mathbb{Z}$ the winding number of γ about 0. In the same way $\int_h^{h+\tau} g = 2\pi i\,\ell$ with $\ell \in \mathbb{Z}$.

Step 6. Combining Steps 3–5, $\sum_i a_i' - \sum_j b_j' = \ell - n\tau \in \Lambda$. ∎

Exercise 9.3 (The exponent 2 is critical)

(a) Show that $\sum_{\omega\in\Lambda^*}|\omega|^{-2}$ diverges for every $\tau \in \mathbb{H}$.
(b) Show that, as $R \to \infty$,

$$\sum_{1\le n^2+m^2\le R^2} \frac{1}{n^2+m^2} = 2\pi\log R + O(1).$$

Solution.

(b) Write $\omega = (n,m) \in \mathbb{Z}^2$, $f(x) = |x|^{-2}$ on $\mathbb{R}^2 \setminus \{0\}$, $c = \sqrt{2}/2$, and let Q_ω be the closed unit square centered at ω. Take $R \ge 4$.

Step 1. (Local comparison.) For $|\omega| \ge 2$, $x \in Q_\omega$, every $y \in [\omega, x] \subset Q_\omega$ has $|y| \ge |\omega|/2$ and $|\nabla f(y)| = 2|y|^{-3} \le 16|\omega|^{-3}$; the segment has length $\le c$, so

$$|f(x) - f(\omega)| \le 8\sqrt{2}\,|\omega|^{-3}.$$

Step 2. (Sum versus integral.) The six terms with $1 \le |\omega| < 2$ ($(\pm1,0)$, $(0,\pm1)$, $(\pm1,\pm1)$) total $4 + 4\cdot\frac{1}{2} = 6$; let $S_1(R)$ be the sum over $2 \le |\omega| \le R$ and U_R the union of the unit squares Q_ω there (disjoint interiors). By Step 1,

$$\Big|S_1(R) - \int_{U_R} f\,dx\Big| \le 8\sqrt{2}\sum_{|\omega|\ge 2}|\omega|^{-3} =: C < \infty,$$

the series converging by Lemma 1.5 with $r = 3$, $\tau = i$.

Step 3. (Annuli.) Let $A(a,b)=\{a\le|x|\le b\}$. If $2+c\le|x|\le R-c$, then x lies in some Q_ω, and $|x-\omega|\le c$ gives $2\le|\omega|\le R$; so $A(2+c,R-c)\subset U_R$. Conversely, $x\in Q_\omega$ with $2\le|\omega|\le R$ has $2-c\le|x|\le R+c$, so $U_R\subset A(2-c,R+c)$. In polar coordinates $\int_{A(a,b)}f=\int_0^{2\pi}\int_a^b r^{-1}\,dr\,d\theta=2\pi\log(b/a)$, and $f>0$, so

$$2\pi\log\frac{R-c}{2+c}\le\int_{U_R}f\,dx\le 2\pi\log\frac{R+c}{2-c}.$$

Step 4. (Conclusion.) For $R\ge 4$, $|\log(1\pm c/R)|\le\log 2$, so both bounds equal $2\pi\log R+O(1)$. With Step 2 the full sum is $6+2\pi\log R+O(1)=2\pi\log R+O(1)$.

(a) With $\tau\in\mathbb{H}$ and $\omega=n+m\tau\neq 0$,

$$|n+m\tau|\le|n|+|m||\tau|\le(1+|\tau|)(|n|+|m|)\le\sqrt{2}(1+|\tau|)(n^2+m^2)^{1/2}.$$

Hence $|\omega|^{-2}\ge\frac{1}{2(1+|\tau|)^2}(n^2+m^2)^{-1}$, and the partial sums over $n^2+m^2\le R^2$ are at least a fixed multiple of $2\pi\log R+O(1)\to\infty$ by (b). So the series diverges. ∎

Exercise 9.7 ($\zeta(2)$ and $\zeta(4)$ from lattice sums)

Using $\sum_{m\in\mathbb{Z}}(m+\tau)^{-2}=\pi^2/\sin^2(\pi\tau)$ at $\tau=1/2$, show

$$\sum_{m\ge1,\ m\text{ odd}}\frac{1}{m^2}=\frac{\pi^2}{8},\qquad\zeta(2)=\frac{\pi^2}{6},$$

and, using $\sum_{m\in\mathbb{Z}}(m+\tau)^{-4}$, show

$$\sum_{m\ge1,\ m\text{ odd}}\frac{1}{m^4}=\frac{\pi^4}{96},\qquad\zeta(4)=\frac{\pi^4}{90}.$$

Solution.

Step 1. (The identities on $\mathbb{C}\setminus\mathbb{Z}$.) By Chapter 5, $\pi\cot\pi z=\lim_N\sum_{|n|\le N}\frac{1}{z+n}$ for $z\notin\mathbb{Z}$, uniformly on compact subsets of $\mathbb{C}\setminus\mathbb{Z}$. Differentiating termwise (Weierstrass), as in the proof of Lemma 2.4,

$$\sum_{n\in\mathbb{Z}}\frac{1}{(z+n)^2}=\frac{\pi^2}{\sin^2\pi z}=:g(z),\qquad z\in\mathbb{C}\setminus\mathbb{Z},$$

and the left side converges absolutely and locally uniformly, so it may be differentiated twice more termwise:

$$\sum_{n\in\mathbb{Z}}\frac{6}{(z+n)^4}=g''(z).$$

Step 2. (Rewriting at $z=1/2$.) $(n+\frac12)^{-k}=2^k(2n+1)^{-k}$, and as n runs over $\mathbb{Z}$, $2n+1$ runs over the odd integers, each odd $m\ge1$ arising twice ($\pm m$). So for even k,

$$\sum_{n\in\mathbb{Z}}\frac{1}{(n+\frac12)^k}=2^{k+1}\sum_{m\ge1\text{ odd}}\frac{1}{m^k}.$$

Step 3. ($k=2$.) $g(1/2)=\pi^2$, so $\sum_{m\text{ odd}}m^{-2}=\pi^2/8$. Splitting $\zeta(2)$ into odd and even terms, $\zeta(2)=\frac{\pi^2}{8}+\frac14\zeta(2)$, hence $\zeta(2)=\pi^2/6$.

Step 4. ($k=4$.) With $u=\pi z$,

$$g'(z)=-2\pi^3\csc^2u\cot u,\qquad g''(z)=2\pi^4(2\csc^2u\cot^2u+\csc^4u).$$

At $z=1/2$, $\csc u=1$ and $\cot u=0$, so $g''(1/2)=2\pi^4$ and $\sum_n(n+\frac12)^{-4}=\pi^4/3$. By Step 2, $\sum_{m\text{ odd}}m^{-4}=\frac{\pi^4}{3\cdot32}=\frac{\pi^4}{96}$. Then $\zeta(4)=\frac{\pi^4}{96}+\frac{1}{16}\zeta(4)$, so $\zeta(4)=\frac{16}{15}\cdot\frac{\pi^4}{96}=\frac{\pi^4}{90}$. ∎

Chapter 10. Applications of Theta Functions

Overview

The *theta function*

$$\Theta(z|\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z}$$

converges for all $z \in \mathbb{C}$, $\tau \in \mathbb{H}$. In z it is elliptic-like (period 1, "quasi-period" τ); in τ it is modular, connected with partitions and sums of squares. The tools are the *triple product formula* and the *transformation law*; applications are partitions and the two- and four-squares theorems.

1 Product formula for the Jacobi theta function

Motivation. $\Theta(z|\tau)$ is entire with explicit zeros, and its product formula yields partition and sums-of-squares identities.

For $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$, **Jacobi's theta function** is

$$\Theta(z|\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z}. \tag{1}$$

Special cases: $\theta(\tau) = \Theta(0|\tau) = \sum_n e^{\pi i n^2 \tau}$, $\vartheta(t) = \theta(it)$; the heat kernel on the circle (Book I) is $H_t(x) = \sum_n e^{-4\pi^2 n^2 t} e^{2\pi i n x} = \Theta(x|4\pi i t)$; ϑ's functional equation gave ζ's (Chapter 6).

Proposition 1.1

The function Θ satisfies the following properties:
(i) Θ is entire in $z \in \mathbb{C}$ and holomorphic in $\tau \in \mathbb{H}$.
(ii) $\Theta(z+1|\tau) = \Theta(z|\tau)$.
(iii) $\Theta(z+\tau|\tau) = \Theta(z|\tau) e^{-\pi i \tau} e^{-2\pi i z}$.
(iv) $\Theta(z|\tau) = 0$ whenever $z = 1/2 + \tau/2 + n + m\tau$ and $n, m \in \mathbb{Z}$.

Proof. If $\mathrm{Im}(\tau) = t \geq t_0 > 0$ and $|z| \leq M$, then $\sum_n |e^{\pi i n^2 \tau} e^{2\pi i n z}| \leq C \sum_{n \geq 0} e^{-\pi n^2 t_0} e^{2\pi n M} < \infty$, uniformly; this gives (i).

(ii) is clear since $e^{2\pi i n z}$ has period 1. For (iii), complete the square:

$$\Theta(z+\tau|\tau) = \sum_n e^{\pi i (n^2+2n)\tau} e^{2\pi i n z} = \sum_n e^{\pi i (n+1)^2 \tau} e^{-\pi i \tau} e^{2\pi i (n+1) z} e^{-2\pi i z} = \Theta(z|\tau) e^{-\pi i \tau} e^{-2\pi i z}.$$

By (ii) and (iii) it suffices for (iv) to show $\Theta(1/2+\tau/2|\tau) = 0$: $\Theta(1/2+\tau/2|\tau) = \sum_n (-1)^n e^{\pi i (n^2+n)\tau}$, and pairing $n \geq 0$ with $-n-1$, which has opposite parity and $(-n-1)^2 + (-n-1) = n^2 + n$, the terms cancel. ∎

With the standard notation $q = e^{\pi i \tau}$, the **triple product** is

$$\Pi(z|\tau) = \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1} e^{2\pi i z})(1 + q^{2n-1} e^{-2\pi i z}).$$

Proposition 1.2

The function $\Pi(z|\tau)$ satisfies the following properties:
(i) $\Pi(z|\tau)$ is entire in $z \in \mathbb{C}$ and holomorphic for $\tau \in \mathbb{H}$.
(ii) $\Pi(z+1|\tau) = \Pi(z|\tau)$.
(iii) $\Pi(z+\tau|\tau) = \Pi(z|\tau)e^{-\pi i\tau}e^{-2\pi iz}$.
(iv) $\Pi(z|\tau) = 0$ whenever $z = 1/2 + \tau/2 + n + m\tau$ and $n, m \in \mathbb{Z}$. Moreover, these points are simple zeros of $\Pi(\cdot|\tau)$, and $\Pi(\cdot|\tau)$ has no other zeros.

Proof. If $\mathrm{Im}(\tau) \geq t_0 > 0$, $|q| \leq e^{-\pi t_0} < 1$ and each factor is $1+O(|q|^{2n-1}e^{2\pi|z|})$; since $\sum|q|^{2n-1} < \infty$, the results on infinite products (Section 3 of Chapter 5) give (i).

(ii) is clear. For (iii), since $q^2 = e^{2\pi i\tau}$,

$$\Pi(z+\tau|\tau) = \prod_{n=1}^{\infty}(1-q^{2n})(1+q^{2n+1}e^{2\pi iz})(1+q^{2n-3}e^{-2\pi iz}),$$

and comparing with $\Pi(z|\tau)$, $\Pi(z+\tau|\tau) = \Pi(z|\tau)\frac{1+q^{-1}e^{-2\pi iz}}{1+qe^{2\pi iz}}$;

(iii) follows from $(1+x)/(1+x^{-1}) = x$. For (iv), a convergent product vanishes only where a factor does, and $1-q^{2n} \neq 0$. $1+q^{2n-1}e^{2\pi iz} = 0$ iff $(2n-1)\tau + 2z \equiv 1 \pmod 2$, i.e. $z \equiv 1/2+\tau/2-n\tau \pmod 1$; $1+q^{2n-1}e^{-2\pi iz} = 0$ iff $z \equiv 1/2+\tau/2+(n-1)\tau \pmod 1$ $(n \geq 1)$. These are all the zeros, simple as $e^w - 1$ has a simple zero at 0. ∎

Theorem 1.3 (Product formula)

For all $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$ we have the identity $\Theta(z|\tau) = \Pi(z|\tau)$.

Proof. Fix $\tau \in \mathbb{H}$. There is a constant $c(\tau)$ with

$$\Theta(z|\tau) = c(\tau)\Pi(z|\tau)\,: \tag{2}$$

$F(z) = \Theta(z|\tau)/\Pi(z|\tau)$ is entire (by (iv) of both propositions) and doubly periodic with periods 1, τ (by (ii), (iii)), hence constant. We show $c(\tau) = c(4\tau)$. Put $z = 1/2$ in (2) ($e^{\pm 2\pi iz} = -1$):

$$\sum_n(-1)^n q^{n^2} = c(\tau)\prod_{n=1}^{\infty}(1-q^{2n})(1-q^{2n-1})^2 = c(\tau)\prod_{n=1}^{\infty}(1-q^n)(1-q^{2n-1}), \quad c(\tau) = \frac{\sum_n(-1)^n q^{n^2}}{\prod_{n=1}^{\infty}(1-q^n)(1-q^{2n-1})}. \tag{3}$$

Put $z = 1/4$ ($e^{2\pi iz} = i$): $\Theta(1/4|\tau) = \sum_n q^{n^2}i^n$, and since $1/i = -i$, only even $n = 2m$ survive: $\Theta(1/4|\tau) = \sum_m q^{4m^2}(-1)^m$. On the other hand

$$\Pi(1/4|\tau) = \prod_{m=1}^{\infty}(1-q^{2m})(1+iq^{2m-1})(1-iq^{2m-1}) = \prod_{m=1}^{\infty}(1-q^{2m})(1+q^{4m-2}) = \prod_{n=1}^{\infty}(1-q^{4n})(1-q^{8n-4}),$$

the last step by splitting $2m = 4n$ and $2m = 4n-2$ in the first factor ($(1-q^{4n-2})(1+q^{4n-2}) = 1-q^{8n-4}$). Hence

$$c(\tau) = \frac{\sum_n(-1)^n q^{4n^2}}{\prod_{n=1}^{\infty}(1-q^{4n})(1-q^{8n-4})}, \tag{4}$$

and comparing (3) and (4) ($q \mapsto q^4$ corresponds to $\tau \mapsto 4\tau$), $c(\tau) = c(4\tau)$. So $c(\tau) = c(4^k\tau)$, and since $q^{4^k} = e^{\pi i 4^k\tau} \to 0$, (3) gives $c(4^k\tau) \to 1$; so $c(\tau) = 1$. ∎

Corollary 1.4

If $\operatorname{Im}(\tau) > 0$ and $q = e^{\pi i \tau}$, then

$$\theta(\tau) = \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1})^2.$$

Thus $\theta(\tau) \neq 0$ for $\tau \in \mathbb{H}$.

Corollary 1.5

For each fixed $\tau \in \mathbb{H}$, the quotient

$$(\log \Theta(z|\tau))'' = \frac{\Theta(z|\tau)\Theta''(z|\tau) - (\Theta'(z|\tau))^2}{\Theta(z|\tau)^2}$$

is an elliptic function of order 2 with periods 1 and τ, and with a double pole at $z = 1/2 + \tau/2$.

(Primes denote differentiation in z.)

Proof. Let $F = \Theta'/\Theta$. Differentiating (ii) and (iii) of Proposition 1.1 gives $F(z+1) = F(z)$, $F(z+\tau) = F(z) - 2\pi i$, so F' is doubly periodic; Θ has only a simple zero at $1/2 + \tau/2$ in the fundamental parallelogram, so F' has only a double pole. ∎

1.1 Further transformation laws

In Chapter 9 the modular character was expressed through $T_1 : \tau \mapsto \tau + 1$ and $S : \tau \mapsto -1/\tau$. For Θ it is natural to use $T_2 : \tau \mapsto \tau + 2$ and S, since $\Theta(z|\tau + 2) = \Theta(z|\tau)$ but $\Theta(z|\tau + 1) \neq \Theta(z|\tau)$.

Theorem 1.6

If $\tau \in \mathbb{H}$, then

$$\Theta(z|-1/\tau) = \sqrt{\tau/i}\, e^{\pi i z^2 \tau} \Theta(z\tau|\tau) \qquad \text{for all } z \in \mathbb{C}. \tag{5}$$

Here $\sqrt{\tau/i}$ denotes the branch of the square root defined on the upper half-plane that is positive when $\tau = it$, $t > 0$.

Proof. It suffices to take $z = x$ real and $\tau = it$: both sides are holomorphic in τ for fixed real x, and entire in z for fixed τ. Then (5) reads

$$\sum_n e^{-\pi n^2/t} e^{2\pi i n x} = t^{1/2} e^{-\pi t x^2} \sum_n e^{-\pi n^2 t} e^{-2\pi n x t}.$$

Replacing x by a, this is $\sum_n e^{-\pi t(n+a)^2} = \sum_n t^{-1/2} e^{-\pi n^2/t} e^{2\pi i n a}$, which is equation (3) of Chapter 4 (from Poisson summation). ∎

Corollary 1.7

If $\operatorname{Im}(\tau) > 0$, then $\theta(-1/\tau) = \sqrt{\tau/i}\,\theta(\tau)$.

(Set $z = 0$; for $\tau = it$ this is the functional equation of ϑ.) This gives precise information about θ as $\tau \to 0$; the next corollary describes θ near $\tau = 1$.

Corollary 1.8

If $\tau \in \mathbb{H}$, then

$$\theta(1-1/\tau) = \sqrt{\tau/i}\sum_{n=-\infty}^{\infty} e^{\pi i(n+1/2)^2\tau} = \sqrt{\tau/i}\,(2e^{\pi i\tau/4}+\cdots).$$

The second identity means that $\theta(1-1/\tau) \sim 2\sqrt{\tau/i}\,e^{\pi i\tau/4}$ as $\mathrm{Im}(\tau)\to\infty$.

Proof. n and n^2 have the same parity, so $\theta(1+\tau) = \sum_n (-1)^n e^{\pi i n^2\tau} = \Theta(1/2|\tau)$; hence $\theta(1-1/\tau) = \Theta(1/2|-1/\tau)$. By Theorem 1.6 with $z = 1/2$,

$$\theta(1-1/\tau) = \sqrt{\tau/i}\,e^{\pi i\tau/4}\Theta(\tau/2|\tau) = \sqrt{\tau/i}\,e^{\pi i\tau/4}\sum_n e^{\pi i n^2\tau}e^{\pi i n\tau} = \sqrt{\tau/i}\sum_n e^{\pi i(n+1/2)^2\tau}.$$

The terms $n = 0, -1$ contribute $2e^{\pi i\tau/4}$, of absolute value $2e^{-\pi t/4}$ ($\tau = \sigma + it$); the others total $O(\sum_{k\ge1} e^{-\pi(k+1/2)^2 t}) = O(e^{-9\pi t/4})$. ∎

The **Dedekind eta function** is, for $\mathrm{Im}(\tau) > 0$,

$$\eta(\tau) = e^{\frac{\pi i\tau}{12}}\prod_{n=1}^{\infty}(1-e^{2\pi i n\tau}).$$

Proposition 1.9

If $\mathrm{Im}(\tau) > 0$, then $\eta(-1/\tau) = \sqrt{\tau/i}\,\eta(\tau)$.

Proof. Differentiate (5) and evaluate at $z_0 = 1/2 + \tau/2$. By the product formula,

$$\Theta(z|\tau) = (1+qe^{-2\pi iz})\prod_{n=1}^{\infty}(1-q^{2n})(1+q^{2n-1}e^{2\pi iz})(1+q^{2n+1}e^{-2\pi iz}),$$

and the first factor vanishes at z_0 (where $e^{2\pi iz_0} = -q$), with derivative $-2\pi iqe^{-2\pi iz_0} = 2\pi i$ there, while the remaining factors equal $(1-q^{2n})^3$. So

$$\Theta'(z_0|\tau) = 2\pi i H(\tau), \qquad H(\tau) = \prod_{n=1}^{\infty}(1-e^{2\pi i n\tau})^3.$$

Replacing $-1/\tau$ by τ in (5), $\Theta(z|\tau) = \sqrt{i/\tau}\,e^{-\pi iz^2/\tau}\Theta(-z/\tau|-1/\tau)$. Differentiating and evaluating at z_0 (where $-z_0/\tau = -\frac{1}{2\tau} - \frac{1}{2}$ is a zero of $\Theta(\cdot|-1/\tau)$, congruent to $\frac{1}{2} + \frac{-1/\tau}{2}$), a computation gives

$$2\pi i H(\tau) = \sqrt{i/\tau}\,e^{-\frac{\pi i\tau}{4}}e^{-\frac{\pi i}{2}}e^{-\frac{\pi i}{4\tau}}\Big(-\frac{1}{\tau}\Big)2\pi i H(-1/\tau), \qquad \text{i.e.} \qquad e^{\frac{\pi i\tau}{4}}H(\tau) = \Big(\frac{i}{\tau}\Big)^{3/2}e^{-\frac{\pi i}{4\tau}}H(-1/\tau),$$

which says $\eta(\tau)^3 = (i/\tau)^{3/2}\eta(-1/\tau)^3$. For $\tau = it$, $\eta(\tau) > 0$, so taking cube roots gives $\eta(\tau) = \sqrt{i/\tau}\,\eta(-1/\tau)$; by analytic continuation this holds on $\mathbb{H}$. ∎

2 Generating functions

The **generating function** of $\{F_n\}_{n\ge0}$ is $F(x) = \sum_n F_n x^n$. The **partition function** $p(n)$ counts the ways to write n as a sum of positive integers ($p(0) = 1$; $p(1) = 1$, $p(2) = 2$, $p(3) = 3$, $p(4) = 5$, $p(5) = 7$, $p(6) = 11$, $p(7) = 15$, $p(8) = 22$, $p(12) = 77$).

Theorem 2.1 (Euler)

If $|x|<1$, then $\displaystyle\sum_{n=0}^{\infty} p(n)x^n = \prod_{k=1}^{\infty} \frac{1}{1-x^k}$.

Formally, expand $\frac{1}{1-x^k} = \sum_m x^{km}$ and multiply: a partition $n = m_1k_1 + \cdots + m_rk_r$ (distinct k_j) gives the term $(x^{k_1})^{m_1}\cdots(x^{k_r})^{m_r}$. This is justified as for the product formula of ζ (Chapter 7, Section 1), since $\frac{1}{1-x^k} = 1 + O(x^k)$.

Similarly $\prod 1/(1-x^{2n-1})$ generates $p_o(n)$, the number of partitions of n into odd parts, and $\prod(1+x^n)$ generates $p_u(n)$, partitions into unequal parts. Remarkably $p_o(n) = p_u(n)$, i.e.

$$\prod_{n=1}^{\infty} \frac{1}{1-x^{2n-1}} = \prod_{n=1}^{\infty}(1+x^n)\,:$$

indeed $\prod(1+x^n)\prod(1-x^n) = \prod(1-x^{2n})$, and by parity $\prod(1-x^{2n})\prod(1-x^{2n-1}) = \prod(1-x^n)$.

Let $p_{e,u}(n)$, $p_{o,u}(n)$ count partitions of n into an even, resp. odd, number of unequal parts. Euler: $p_{e,u}(n) = p_{o,u}(n)$ unless $n = k(3k+1)/2$, $k \in \mathbb{Z}$, is **pentagonal** $(1, 2, 5, 7, 12, 15, 22, 26, \dots)$, when $p_{e,u}(n) - p_{o,u}(n) = (-1)^k$. Multiplying out gives terms $(-1)^r x^{n_1+\cdots+n_r}$ with distinct n_j, so

$$\prod_{n=1}^{\infty}(1-x^n) = \sum_{n=0}^{\infty}[p_{e,u}(n) - p_{o,u}(n)]x^n,$$

so Euler's theorem follows from:

Proposition 2.2 (Euler's pentagonal number theorem)

$$\prod_{n=1}^{\infty}(1-x^n) = \sum_{k=-\infty}^{\infty}(-1)^k x^{\frac{k(3k+1)}{2}}.$$

Proof. Set $x = e^{2\pi iu}$. In the triple product take $q = e^{3\pi iu}$ and $z = 1/2 + u/2$; then

$$\prod_{n=1}^{\infty}(1-e^{2\pi i3nu})(1-e^{2\pi i(3n-1)u})(1-e^{2\pi i(3n-2)u}) = \prod_{n=1}^{\infty}(1-e^{2\pi inu}).$$

By Theorem 1.3 the product equals $\sum_n e^{3\pi in^2u}(-1)^n e^{2\pi inu/2} = \sum_n (-1)^n e^{\pi in(3n+1)u} = \sum_n (-1)^n x^{n(3n+1)/2}$. ∎

The growth of $p(n)$ follows from $1/\prod(1-x^n)$ as $|x| \to 1$; the transformation law of η (Proposition 1.9) gives a precise asymptotic formula (Appendix A).

3 The theorems about sums of squares

$r_2(n)$ counts $(x,y) \in \mathbb{Z}^2$ with $n = x^2+y^2$ (e.g. $r_2(3) = 0$, $r_2(5) = 8$), $r_4(n)$ the representations as sums of four squares. No $4k+3$ is a sum of two squares, no $8k+7$ of three. Which n are sums of two squares, is every n a sum of four, and what are r_2, r_4? Jacobi's theta functions first answered these fully.

3.1 The two-squares theorem

If $n = a^2+b^2$ and $m = c^2+d^2$, then with $x+iy = (a+ib)(c+id)$, $x^2+y^2 = nm$; so divisibility properties of n matter. Let $d_1(n)$ and $d_3(n)$ be the numbers of divisors of n of the form $4k+1$ and $4k+3$.

Theorem 3.1 (Two-squares theorem)

If $n \geq 1$, then $r_2(n) = 4(d_1(n) - d_3(n))$.

So $n = p_1^{a_1} \cdots p_r^{a_r}$ is a sum of two squares iff every $p_j \equiv 3 \pmod 4$ has even a_j. For $q = e^{\pi i \tau}$, $\tau \in \mathbb{H}$,

$$\theta(\tau)^2 = \sum_{n=0}^{\infty} r_2(n) q^n, \tag{6}$$

since $\theta(\tau)^2 = \sum_{n_1} \sum_{n_2} q^{n_1^2} q^{n_2^2} = \sum_{(n_1,n_2)\in\mathbb{Z}^2} q^{n_1^2+n_2^2}$.

Proposition 3.2

The identity $r_2(n) = 4(d_1(n) - d_3(n))$, $n \geq 1$, is equivalent to the identities

$$\theta(\tau)^2 = 2 \sum_{n=-\infty}^{\infty} \frac{1}{q^n + q^{-n}} = 1 + 4 \sum_{n=1}^{\infty} \frac{q^n}{1+q^{2n}}, \tag{7}$$

whenever $q = e^{\pi i \tau}$ and $\tau \in \mathbb{H}$.

Proof. Both series converge absolutely ($|q| < 1$), and they are equal since $1/(q^n + q^{-n}) = q^{|n|}/(1 + q^{2|n|})$. Since $(1+q^{2n})^{-1} = (1-q^{2n})/(1-q^{4n})$, the right side of (7) is $1 + 4\sum_{n\geq 1}\left[\frac{q^n}{1-q^{4n}} - \frac{q^{3n}}{1-q^{4n}}\right]$. Now $\frac{1}{1-q^{4n}} = \sum_{m\geq 0} q^{4nm}$, so

$$\sum_{n=1}^{\infty} \frac{q^n}{1-q^{4n}} = \sum_{n=1}^{\infty}\sum_{m=0}^{\infty} q^{n(4m+1)} = \sum_{k=1}^{\infty} d_1(k) q^k, \qquad \sum_{n=1}^{\infty} \frac{q^{3n}}{1-q^{4n}} = \sum_{k=1}^{\infty} d_3(k) q^k$$

(the series converge since $d_j(k) \leq k$). Compare with (6). ∎

Let

$$C(\tau) = 2 \sum_{n=-\infty}^{\infty} \frac{1}{q^n + q^{-n}} = \sum_{n=-\infty}^{\infty} \frac{1}{\cos(\pi n \tau)}, \qquad q = e^{\pi i \tau},\ \tau \in \mathbb{H}. \tag{8}$$

We must prove $\theta^2 = C$. In parallel: θ comes from the heat kernel $e^{-\pi x^2}$ on the line, C from the strip kernel $1/\cosh \pi x$ (Section 1), both their own Fourier transforms (Chapter 3, Section 2, Example 3), and both transformation laws come from Poisson summation. Indeed $\theta(\tau)^2 = (i/\tau)\theta(-1/\tau)^2$ (Corollary 1.7), and (5) of Chapter 4 with $a = 0$ gives $\sum_n \frac{1}{\cosh \pi n t} = \frac{1}{t}\sum_n \frac{1}{\cosh(\pi n/t)}$, i.e. $C(\tau) = (i/\tau)C(-1/\tau)$ on $\tau = it$, hence on $\mathbb{H}$. Both tend to 1 as $\mathrm{Im}(\tau) \to \infty$. At the cusp $\tau = 1$: $\theta(1-1/\tau)^2 \sim 4(\tau/i)e^{\pi i \tau/2}$ (Corollary 1.8), and (5) of Chapter 4 with $a = 1/2$ gives $\sum_n \frac{(-1)^n}{\cosh(\pi n/t)} = t\sum_n \frac{1}{\cosh(\pi(n+1/2)t)}$, so by analytic continuation

$$C(1-1/\tau) = \frac{\tau}{i} \sum_n \frac{1}{\cos(\pi(n+1/2)\tau)} = 4\frac{\tau}{i} e^{\pi i \tau/2} + O(|\tau| e^{-3\pi t/2}) \quad (t \to \infty),$$

the main terms being $n = 0, -1$.

Proposition 3.3

The function $C(\tau) = \sum 1/\cos(\pi n \tau)$, defined in the upper half-plane, satisfies (i) $C(\tau + 2) = C(\tau)$.
(ii) $C(\tau) = (i/\tau)C(-1/\tau)$.
(iii) $C(\tau) \to 1$ as $\mathrm{Im}(\tau) \to \infty$.
(iv) $C(1-1/\tau) \sim 4(\tau/i)e^{\pi i \tau/2}$ as $\mathrm{Im}(\tau) \to \infty$. Moreover, $\theta(\tau)^2$ satisfies the same properties.

Theorem 3.4

Suppose f is a holomorphic function in the upper half-plane that satisfies:
(i) $f(\tau+2)=f(\tau)$, (ii) $f(-1/\tau)=f(\tau)$, (iii) $f(\tau)$ is bounded, then f is constant.

Let $\mathcal{F}=\{\tau\in\overline{\mathbb{H}} : |\mathrm{Re}(\tau)|\le 1,\ |\tau|\ge 1\}$; the points $\tau=\pm1$ are its **cusps** (equivalent under $\tau\mapsto\tau+2$).

Lemma 3.5

Every point of $\mathbb{H}$ maps into $\mathcal{F}$ under a composite of $T_2:\tau\mapsto\tau+2$, $S:\tau\mapsto -1/\tau$ and their inverses.

$\mathcal{F}$ is the **fundamental domain** of the group G generated by T_2, S (representatives unique except on $\partial\mathcal{F}$). Elements $g\in G$ are matrices $\begin{pmatrix} a & b \\ c & d\end{pmatrix}$, $g(\tau)=\frac{a\tau+b}{c\tau+d}$, with integer entries and determinant 1, so

$$\mathrm{Im}(g(\tau))=\frac{\mathrm{Im}(\tau)}{|c\tau+d|^2}. \tag{9}$$

Proof. By (9), with c,d integers, some $g_0\in G$ maximizes $\mathrm{Im}(g_0(\tau))$; $T_2^{\pm1}$ preserve imaginary parts, so some g_1 has $|\mathrm{Re}(g_1(\tau))|\le 1$ and maximal Im. If $|g_1(\tau)|<1$, $\mathrm{Im}(Sg_1(\tau))=\mathrm{Im}(g_1(\tau))/|g_1(\tau)|^2$ is larger, a contradiction. So $g_1(\tau)\in\mathcal{F}$. ■

Proof of Theorem 3.4. Suppose f is not constant. $g(z)=f(\tau)$, $z=e^{\pi i\tau}$, is well defined on the punctured disc (period 2) and bounded, so $\lim_{\mathrm{Im}\,\tau\to\infty} f(\tau)$ exists, with modulus $<\sup_{\mathcal{F}}|f|$ by maximum modulus. At the cusps (by $f(\tau+2)=f(\tau)$, only $\tau=1$) the same holds for $\lim f(1-1/\tau)$, by interchanging 1 and ∞: let U_n correspond to $\begin{pmatrix} 1-n & n \\ -n & 1+n\end{pmatrix}$, fixing 1; with $\nu(\tau)=1/(1-\tau)$, $\nu^{-1}(\tau)=1-1/\tau$, $U_n=\nu^{-1}T_n\nu$, $T_n(\tau)=\tau+n$. So $U_nU_m=U_{n+m}$ and $U_{-1}=\begin{pmatrix} 2 & -1 \\ 1 & 0\end{pmatrix}=T_2S$, every U_n lies in G, and $F(\tau)=f(1-1/\tau)$ has period 1; $h(z)=F(\tau)$, $z=e^{2\pi i\tau}$, has a removable singularity at 0. So f attains its maximum inside $\mathbb{H}$, contradicting the maximum principle. ■

Proof of the two-squares theorem. $f=C/\theta^2$ is holomorphic in $\mathbb{H}$ ($\theta\neq 0$, Corollary 1.4), invariant under T_2 and S, and in $\mathcal{F}$ bounded with limit 1 as $\mathrm{Im}(\tau)\to\infty$ or $\tau\to\pm1$, by (iii) and (iv) of Proposition 3.3. So $f\equiv 1$ by Theorem 3.4: $\theta^2=C$. ■

3.2 The four-squares theorem

Let $\sigma_1^*(n)$ be the sum of the divisors of n that are not divisible by 4.

Theorem 3.6 (Four-squares theorem)

Every positive integer is the sum of four squares, and moreover $r_4(n)=8\sigma_1^*(n)$ for all $n\ge 1$.

As before $\theta(\tau)^4=\sum_{n\ge0} r_4(n)q^n$, $q=e^{\pi i\tau}$, but nothing as simple as C is available; we use a variant of the forbidden Eisenstein series $F(\tau)=\sum_m\sum_n \frac{1}{(n+m\tau)^2}$ (in n, then m; $(n,m)\neq(0,0)$) of Corollary 2.6, defining

$$E_2^*(\tau)=F(\tau/2)-4F(2\tau),\qquad \tau\in\mathbb{H}, \tag{10}$$

equivalently $E_2^*(\tau)=\sum_m\sum_n\left[\frac{1}{(n+m\tau/2)^2}-\frac{1}{(n/2+m\tau)^2}\right]$; the order of summation is critical.

Proposition 3.7

The assertion $r_4(n) = 8\sigma_1^*(n)$ is equivalent to the identity $\theta(\tau)^4 = -\pi^{-2}E_2^*(\tau)$, where $\tau \in \mathbb{H}$.

Proof. It suffices to show $-\pi^{-2}E_2^*(\tau) = 1 + 8\sum_{k\geq 1}\sigma_1^*(k)q^k$. By Corollary 2.6, $F(\tau) = \frac{\pi^2}{3} - 8\pi^2\sum_{k\geq 1}\sigma_1(k)e^{2\pi ik\tau}$. And $\sigma_1^*(n) = \sigma_1(n)$ if $4 \nmid n$, $\sigma_1(n) - 4\sigma_1(n/4)$ if $4 \mid n$ (divisors $d = 4\tilde{d}$ of $n = 4\tilde{n}$ match divisors $\tilde{d}$ of $\tilde{n}$). With (10),

$$E_2^*(\tau) = -\pi^2 - 8\pi^2\sum_{k=1}^{\infty}\sigma_1^*(k)e^{\pi ik\tau}.$$

■

So Theorem 3.6 reduces to $\theta^4 = -\pi^{-2}E_2^*$, and the key is that E_2^* has the same modular properties as θ^4.

Proposition 3.8

The function $E_2^*(\tau)$ defined in the upper half-plane has the following properties:
(i) $E_2^*(\tau+2) = E_2^*(\tau)$.
(ii) $E_2^*(\tau) = -\tau^{-2}E_2^*(-1/\tau)$.
(iii) $E_2^*(\tau) \to -\pi^2$ as $\mathrm{Im}(\tau) \to \infty$.
(iv) $|E_2^*(1-1/\tau)| = O(|\tau^2 e^{\pi i\tau}|)$ as $\mathrm{Im}(\tau) \to \infty$. Moreover $-\pi^2\theta^4$ has the same properties.

(i) is immediate from the definition. Consider F and its *reverse* $\tilde{F}(\tau) = \sum_n\sum_m \frac{1}{(n+m\tau)^2}$ (summing first in m).

Lemma 3.9

The functions F and $\tilde{F}$ satisfy:
(a) $F(-1/\tau) = \tau^2\tilde{F}(\tau)$, (b) $F(\tau) - \tilde{F}(\tau) = 2\pi i/\tau$, (c) $F(-1/\tau) = \tau^2F(\tau) - 2\pi i\tau$.

Proof. (a) follows from the definitions and $(n+m(-1/\tau))^2 = \tau^{-2}(-m+n\tau)^2$. For (b) use $\eta(-1/\tau) = \sqrt{\tau/i}\,\eta(\tau)$ with $\eta(\tau) = q^{1/12}\prod(1-q^{2n})$, $q = e^{\pi i\tau}$. By Proposition 3.2 of Chapter 5, the logarithmic derivative in τ is

$$\frac{\eta'}{\eta}(\tau) = \frac{\pi i}{12} - 2\pi i\sum_{n=1}^{\infty}\frac{nq^{2n}}{1-q^{2n}}, \qquad \sum_{n=1}^{\infty}\frac{nq^{2n}}{1-q^{2n}} = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty}nq^{2nm} = \sum_{k=1}^{\infty}\sigma_1(k)q^{2k}.$$

Since $F(\tau) = \frac{\pi^2}{3} - 8\pi^2\sum_k\sigma_1(k)q^{2k}$, $\frac{\eta'}{\eta}(\tau) = \frac{i}{4\pi}F(\tau)$. By the chain rule the logarithmic derivative of $\eta(-1/\tau)$ is $\tau^{-2}\frac{\eta'}{\eta}(-1/\tau)$, which by (a) equals $\frac{i}{4\pi}\tilde{F}(\tau)$. Taking logarithmic derivatives in the functional equation for η,

$$\frac{i}{4\pi}\tilde{F}(\tau) = \frac{1}{2\tau} + \frac{i}{4\pi}F(\tau),$$

so $\tilde{F}(\tau) = F(\tau) - 2\pi i/\tau$.

(c) follows from (a) and (b). ■

Proof of Proposition 3.8. (ii) By (10) and Lemma 3.9(c),

$$\begin{aligned}E_2^*(-1/\tau) &= F(-1/(2\tau)) - 4F(-2/\tau) = [4\tau^2F(2\tau) - 4\pi i\tau] - 4[(\tau/2)^2F(\tau/2) - \pi i\tau]\\ &= 4\tau^2F(2\tau) - \tau^2F(\tau/2) = -\tau^2(F(\tau/2) - 4F(2\tau)) = -\tau^2E_2^*(\tau).\end{aligned}$$

(iii) In $F(\tau) = \frac{\pi^2}{3} - 8\pi^2\sum\sigma_1(k)e^{2\pi ik\tau}$ the sum tends to 0 as $\mathrm{Im}(\tau) \to \infty$, so by (10) $E_2^*(\tau) \to \frac{\pi^2}{3} - 4\frac{\pi^2}{3} = -\pi^2$.

(iv) We first show

$$E_2^*(1-1/\tau) = \tau^2\Big[F\Big(\frac{\tau-1}{2}\Big) - F(\tau/2)\Big]. \tag{11}$$

F has period 1, and Lemma 3.9(c) can be written $F(w) = w^{-2}[F(-1/w) + 2\pi i w]$. Let $u = (\tau-1)/2$ and $w = u/\tau = \frac{1}{2} - \frac{1}{2\tau}$. Then $-1/w = -2 - \frac{2}{\tau-1}$, so $F(-1/w) = F(-1/u) = u^2F(u) - 2\pi i u$, and

$$F\Big(\frac{1}{2} - \frac{1}{2\tau}\Big) = \frac{\tau^2}{u^2}\big[u^2F(u) - 2\pi i u + 2\pi i u/\tau\big] = \tau^2F(u) - \frac{2\pi i\tau^2(1-1/\tau)}{u} = \tau^2F\Big(\frac{\tau-1}{2}\Big) - 4\pi i\tau.$$

Also $4F(2-2/\tau) = 4F(-1/(\tau/2)) = 4[(\tau/2)^2F(\tau/2) - 2\pi i\tau/2] = \tau^2F(\tau/2) - 4\pi i\tau$. Subtracting gives (11), since $E_2^*(1-1/\tau) = F(\frac{1}{2} - \frac{1}{2\tau}) - 4F(2-\frac{2}{\tau})$. By the expansion of F, $F(\frac{\tau-1}{2}) - F(\tau/2) = -8\pi^2\sum_k \sigma_1(k)e^{\pi ik\tau}((-1)^k - 1) = 16\pi^2e^{\pi i\tau} + O(|e^{3\pi i\tau}|)$, which with (11) gives (iv). For θ^4: (i)–(iii) follow from Section 1 ($\theta(-1/\tau)^4 = (\tau/i)^2\theta(\tau)^4 = -\tau^2\theta(\tau)^4$, $\theta \to 1$), and (iv) from Corollary 1.8: $\theta(1-1/\tau)^4 \sim 16(\tau/i)^2e^{\pi i\tau}$. ∎

Proof of the four-squares theorem. $f = E_2^*/\theta^4$ is holomorphic in $\mathbb{H}$ and invariant under T_2, S (Proposition 3.8). As $\theta(\tau)^4 \to 1$ and $\theta(1-1/\tau)^4 \sim -16\tau^2e^{\pi i\tau}$, f is bounded in $\mathcal{F}$, hence in $\mathbb{H}$, so constant (Theorem 3.4), equal to $-\pi^2$ by (iii). So $\theta^4 = -\pi^{-2}E_2^*$; in particular $r_4(n) \geq 8 > 0$ for $n \geq 1$. ∎

Appendix A. Asymptotics

Overview

Complex analysis helps find asymptotics of functions not computable explicitly. Typically, for large s,

$$I(s) = \int_a^b e^{-s\Phi(x)}\,dx. \tag{1}$$

Three guiding principles:

(i) *Deformation of contour.* For complex Φ the integrand oscillates and cancellations hide the behavior; for holomorphic Φ, move to a contour where Φ is essentially real (Bessel functions).

(ii) *Laplace's method.* For real Φ and $s > 0$ the main contribution comes from near a minimum of Φ (Stirling's formula, the Airy function).

(iii) *Generating functions.* The asymptotics of a sequence F_n can be studied via

$$F_n = \int_\gamma F(e^{2\pi i z})e^{-2\pi i n z}\,dz,$$

γ a segment of unit length in the upper half-plane, a variant of (1); we obtain the asymptotics of the partition function $p(n)$.

1 Bessel functions

Bessel functions arise in problems with rotational symmetry (e.g. the Fourier transform of a radial function on $\mathbb{R}^d$ involves $J_{d/2-1}$). For $\nu > -1/2$,

$$J_\nu(s) = \frac{(s/2)^\nu}{\Gamma(\nu+1/2)\Gamma(1/2)} \int_{-1}^1 e^{isx}(1-x^2)^{\nu-1/2}\,dx. \tag{2}$$

$J_{-1/2}(s) = \lim_{\nu\to -1/2} J_\nu(s) = \sqrt{2/\pi s}\cos s$ and $J_{1/2}(s) = \sqrt{2/\pi s}\sin s$; J_ν is elementary only for half-integral ν. These examples suggest:

Theorem 1.1

$$J_\nu(s) = \sqrt{\frac{2}{\pi s}}\cos\left(s - \frac{\pi\nu}{2} - \frac{\pi}{4}\right) + O(s^{-3/2}) \text{ as } s\to\infty.$$

It suffices to study

$$I(s) = \int_{-1}^1 e^{isx}(1-x^2)^{\nu-1/2}\,dx. \tag{3}$$

Let $f(z) = e^{isz}(1-z^2)^{\nu-1/2}$ on the plane slit along $(-\infty,-1]\cup[1,\infty)$, positive on $(-1,1)$. For fixed $s>0$, Cauchy's theorem on the rectangle $\gamma_{\epsilon,R}$ with vertices $-1+\epsilon$, $1-\epsilon$, $1-\epsilon+iR$, $-1+\epsilon+iR$ ($\epsilon\to 0$, $R\to\infty$) gives

$$I(s) = -I_-(s) - I_+(s),$$

$I_\pm$ the integrals over the vertical half-lines from ± 1 upward, oriented upward for I_+ and downward for I_- appropriately. On the contour of I_+, $z = 1+iy$:

$$I_+(s) = ie^{is}\int_0^\infty e^{-sy}(1-(1+iy)^2)^{\nu-1/2}\,dy, \tag{4}$$

and similarly for I_-. In (3), e^{isx} oscillates rapidly; in (4), e^{-sy} decreases rapidly except near $y=0$, so the main contribution comes from y near 0:

Proposition 1.2

Suppose a and m are fixed, with $a > 0$ and $m > -1$. Then as $s \to \infty$

$$\int_0^a e^{-sx} x^m \, dx = s^{-m-1}\Gamma(m+1) + O(e^{-cs}), \tag{5}$$

for some positive c.

Proof. $m > -1$ guarantees convergence at 0. Write $\int_0^a = \int_0^\infty - \int_a^\infty$. The first equals $s^{-m-1}\Gamma(m+1)$ (change variables $x \mapsto x/s$). For the second,

$$\int_a^\infty e^{-sx} x^m \, dx = e^{-cs} \int_a^\infty e^{-s(x-c)} x^m \, dx = O(e^{-cs}) \tag{6}$$

whenever $c < a$. ∎

Proof of Theorem 1.1. $(1-(1+iy)^2)^{\nu-1/2} = (-2iy)^{\nu-1/2} + O(y^{\nu+1/2})$ for $0 \le y \le 1$, and $= O(y^{\nu-1/2} + y^{2\nu-1})$ for $y \ge 1$. Proposition 1.2 with $a = 1$, $m = \nu - 1/2$ (and $\nu + 1/2$), together with (6), gives

$$I_+(s) = i(-2i)^{\nu-1/2} e^{is} s^{-\nu-1/2}\Gamma(\nu+1/2) + O(s^{-\nu-3/2}),$$
$$I_-(s) = i(2i)^{\nu-1/2} e^{-is} s^{-\nu-1/2}\Gamma(\nu+1/2) + O(s^{-\nu-3/2})$$

(with the appropriate signs). As $J_\nu(s) = \frac{(s/2)^\nu}{\Gamma(\nu+1/2)\Gamma(1/2)}[-I_-(s) - I_+(s)]$, $\Gamma(1/2) = \sqrt{\pi}$, the conjugate terms $(\mp 2i)^{\nu-1/2}$ give phases $e^{\mp i\pi(\nu-1/2)/2}$, and the sum is $2^{\nu+1/2} s^{-\nu-1/2}\Gamma(\nu+\frac{1}{2})\cos(s - \frac{\pi\nu}{2} - \frac{\pi}{4})$ up to $O(s^{-\nu-3/2})$. ∎

Under restricted circumstances Proposition 1.2 extends to $\mathrm{Re}(s) \ge 0$:

Proposition 1.3

Suppose a and m are fixed, with $a > 0$ and $-1 < m < 0$. Then as $|s| \to \infty$ with $\mathrm{Re}(s) \ge 0$,

$$\int_0^a e^{-sx} x^m \, dx = s^{-m-1}\Gamma(m+1) + O(1/|s|).$$

(Here s^{-m-1} is the branch that is positive for $s > 0$.)

Proof. First, for $\mathrm{Re}(s) \ge 0$, $s \ne 0$, $\int_0^\infty e^{-sx} x^m \, dx = \lim_{N\to\infty} \int_0^N e^{-sx} x^m \, dx$ exists and equals $s^{-m-1}\Gamma(m+1)$. Write $\int_0^N = \int_0^a + \int_a^N$; the first is entire in s. Since $-\frac{1}{s}\frac{d}{dx}(e^{-sx}) = e^{-sx}$, integration by parts gives

$$\int_a^N e^{-sx} x^m \, dx = -\frac{m}{s}\int_a^N e^{-sx} x^{m-1} \, dx - \Big[\frac{e^{-sx}x^m}{s}\Big]_a^N. \tag{7}$$

As $m < 0$, $\int_a^\infty x^{m-1} dx < \infty$, so letting $N \to \infty$, $\int_a^\infty e^{-sx} x^m \, dx$ is analytic in $\mathrm{Re}(s) > 0$ and continuous on $\mathrm{Re}(s) \ge 0$, $s \ne 0$; it equals $s^{-m-1}\Gamma(m+1) - \int_0^a$ for $s > 0$, hence everywhere there. Finally $\int_0^a = \int_0^\infty - \int_a^\infty$, and by (7), $\int_a^\infty e^{-sx} x^m \, dx = O(1/|s|)$. ∎

(Better error terms, or other ranges of m, require smooth cut-offs to mitigate the end-point $x = a$.)

2 Laplace's method; Stirling's formula

Motivation. $\int e^{-s\Phi(x)}\psi(x)dx$ for large s concentrates near the minimum of Φ; for $\Gamma(s)$ this gives Stirling.

Consider $\int_a^b e^{-s\Phi(x)}\psi(x)\,dx$ with real Φ, and Φ, ψ indefinitely differentiable. Assume there is $x_0 \in (a,b)$ with $\Phi'(x_0) = 0$ and $\Phi''(x) > 0$ throughout $[a,b]$.

Proposition 2.1 (Laplace's method)

Under the above assumptions, with $s > 0$ and $s \to \infty$,

$$\int_a^b e^{-s\Phi(x)}\psi(x)\,dx = e^{-s\Phi(x_0)}\Big[\frac{A}{s^{1/2}} + O\Big(\frac{1}{s}\Big)\Big], \qquad A = \frac{(2\pi)^{1/2}\psi(x_0)}{(\Phi''(x_0))^{1/2}}. \tag{8}$$

Proof. Replacing Φ by $\Phi - \Phi(x_0)$, assume $\Phi(x_0) = 0$. Since $\Phi'(x_0) = 0$, $\frac{\Phi(x)}{(x-x_0)^2} = \frac{\Phi''(x_0)}{2}\varphi(x)$ with φ smooth, $\varphi(x) = 1 + O(x - x_0)$. Near x_0 change variables $y = (x - x_0)(\varphi(x))^{1/2}$; then $dx/dy = 1 + O(y)$ and $\psi(x) = \tilde{\psi}(y) = \psi(x_0) + O(y)$. On a small interval $[a', b'] \ni x_0$,

$$\int_{a'}^{b'} e^{-s\Phi(x)}\psi(x)\,dx = \psi(x_0)\int_\alpha^\beta e^{-s\frac{\Phi''(x_0)}{2}y^2}\,dy + O\Big(\int_\alpha^\beta e^{-s\frac{\Phi''(x_0)}{2}y^2}|y|\,dy\Big), \tag{9}$$

$\alpha < 0 < \beta$. With $y^2 = X$, $dy = \frac{1}{2}X^{-1/2}dX$, and (5), the first integral is $s^{-1/2}(\frac{2\pi}{\Phi''(x_0)})^{1/2} + O(e^{-\delta s})$; similarly the second is $O(1/s)$. On $[a, a']$ and $[b', b]$, $\Phi \geq c > 0$, so those integrals decay exponentially. ∎

(8) extends to complex s with $\mathrm{Re}(s) \geq 0$, using integration by parts to handle the oscillations of $e^{-s\Phi}$:

Proposition 2.2

With the same assumptions on Φ and ψ, the relation (8) continues to hold if $|s| \to \infty$ with $\mathrm{Re}(s) \geq 0$.

Proof. Proceed to (9); the first term is handled by Proposition 1.3 with $m = -1/2$. Observation: if $\Phi \geq 0$ and $|\Phi'(x)| \geq c > 0$ on $[a,b]$, then for $\mathrm{Re}(s) \geq 0$,

$$\int_a^b e^{-s\Phi(x)}\psi(x)\,dx = O\Big(\frac{1}{|s|}\Big) \qquad \text{as } |s| \to \infty; \tag{10}$$

indeed the integral is $-\frac{1}{s}\int_a^b \frac{d}{dx}(e^{-s\Phi})\frac{\psi}{\Phi'}\,dx = \frac{1}{s}\int_a^b e^{-s\Phi}\frac{d}{dx}\Big(\frac{\psi}{\Phi'}\Big)dx - \frac{1}{s}\Big[e^{-s\Phi}\frac{\psi}{\Phi'}\Big]_a^b$ with $|e^{-s\Phi}| \leq 1$; this handles $[a, a']$, $[b', b]$, where $|\Phi'| \geq c > 0$. The second term of (9) is $\int e^{-s\frac{\Phi''(x_0)}{2}y^2} y\eta(y)\,dy = \frac{-1}{s\Phi''(x_0)}\int \frac{d}{dy}(e^{-s\frac{\Phi''(x_0)}{2}y^2})\eta(y)\,dy$, $O(1/|s|)$ by parts. ∎

For purely imaginary $s = it$ this is the **method of stationary phase**; points with $\Phi'(x_0) = 0$ are **critical points**.

For $\delta > 0$ let $S_\delta = \{s : |\arg s| \leq \pi - \delta\}$, and let $\log s$ be the principal branch.

Theorem 2.3 (Stirling's formula)

If $|s| \to \infty$ with $s \in S_\delta$, then

$$\Gamma(s) = e^{s\log s}e^{-s}\frac{\sqrt{2\pi}}{s^{1/2}}\Big(1 + O\Big(\frac{1}{|s|^{1/2}}\Big)\Big). \tag{11}$$

(With more effort the error term improves to $O(1/|s|)$, with a full asymptotic expansion in powers of $1/s$. Note $\Gamma(s) \sim \sqrt{2\pi}s^{s-1/2}e^{-s}$, the usual form.)

Proof. First let $\mathrm{Re}(s) \geq 0$, with uniform error terms off a neighborhood of 0. For $s > 0$,

$$\Gamma(s) = \int_0^\infty e^{-x} x^s \frac{dx}{x} = \int_0^\infty e^{-x+s\log x} \frac{dx}{x} = e^{s\log s} e^{-s} \int_0^\infty e^{-s\Phi(x)} \frac{dx}{x}, \qquad \Phi(x) = x - 1 - \log x,$$

after $x \mapsto sx$; by analytic continuation $\Gamma(s) = e^{s\log s} e^{-s} I(s)$ for $\mathrm{Re}(s) > 0$, $I(s) = \int_0^\infty e^{-s\Phi(x)} \frac{dx}{x}$. It suffices to show

$$I(s) = \frac{\sqrt{2\pi}}{s^{1/2}} + O\Big(\frac{1}{|s|}\Big) \qquad \text{for } \mathrm{Re}(s) \geq 0. \tag{12}$$

$\Phi(1) = \Phi'(1) = 0$, $\Phi''(x) = 1/x^2 > 0$, $\Phi''(1) = 1$; so Φ is convex, positive, with minimum at 1. Apply Proposition 2.2 on $[1/2, 2]$ with $x_0 = 1$, $\psi(x) = 1/x$: this gives (12) there. On $[0, 1/2]$, since $\Phi'(x) = 1 - 1/x$,

$$\int_\epsilon^{1/2} e^{-s\Phi(x)} \frac{dx}{x} = -\frac{1}{s} \int_\epsilon^{1/2} \frac{d}{dx}(e^{-s\Phi(x)}) \frac{dx}{\Phi'(x)x} = -\frac{1}{s}\Big[\frac{e^{-s\Phi(x)}}{x-1}\Big]_\epsilon^{1/2} - \frac{1}{s}\int_\epsilon^{1/2} e^{-s\Phi(x)} \frac{dx}{(x-1)^2}.$$

Since $\Phi(\epsilon) \to +\infty$ as $\epsilon \to 0$ and $|e^{-s\Phi}| \leq 1$, in the limit

$$\int_0^{1/2} e^{-s\Phi} \frac{dx}{x} = \frac{2}{s} e^{-s\Phi(1/2)} - \frac{1}{s}\int_0^{1/2} e^{-s\Phi} \frac{dx}{(x-1)^2} = O(1/|s|).$$

$\int_2^\infty$ is treated analogously, since $\int_2^\infty (x-1)^{-2} dx < \infty$. So (11) holds for $\mathrm{Re}(s) \geq 0$, $|s| \to \infty$.

To pass to $\mathrm{Re}(s) \leq 0$, $s \in S_\delta$: for $\mathrm{Re}(s) \geq 0$, $s = \sigma + it$, $t \neq 0$, $\log(-s) = \log s - i\pi$ if $t > 0$ and $\log s + i\pi$ if $t < 0$. Hence for $G(s) = e^{s\log s} e^{-s}$,

$$G(-s)^{-1} = e^{s\log s} e^{-s} e^{-si\pi} \ (t > 0), \qquad G(-s)^{-1} = e^{s\log s} e^{-s} e^{si\pi} \ (t < 0). \tag{13}$$

Also

$$\Gamma(s)\Gamma(-s) = \frac{-\pi}{s \sin \pi s}, \tag{14}$$

from $\Gamma(s)\Gamma(1-s) = \pi/\sin \pi s$ and $\Gamma(1-s) = -s\Gamma(-s)$ (Chapter 6, Theorem 1.4, Lemma 1.2). With (13), (14) and $(1 + O(|s|^{-1/2}))^{-1} = 1 + O(|s|^{-1/2})$ this extends (11) to S_δ ($\sin \pi s$ is comparable to $e^{\pm i\pi s}/(\mp 2i)$ for large $|t|$, and S_δ avoids the negative real axis). ∎

3 The Airy function

The Airy function appeared in optics (intensity of light near a caustic):

$$\mathrm{Ai}(s) = \frac{1}{2\pi} \int_{-\infty}^\infty e^{i(x^3/3 + sx)} \, dx, \qquad s \in \mathbb{R}. \tag{15}$$

The integral converges and is continuous in s: $\frac{1}{i(x^2+s)} \frac{d}{dx} e^{i(x^3/3+sx)} = e^{i(x^3/3+sx)}$, so for $a \geq 2|s|^{1/2}$,

$$\int_a^R e^{i(x^3/3+sx)} \, dx = \int_a^R \frac{1}{i(x^2+s)} \frac{d}{dx} e^{i(x^3/3+sx)} \, dx, \tag{16}$$

and integration by parts with $R \to \infty$ gives uniform convergence for $|s| \leq a^2/4$; similarly on $(-\infty, -a]$. Better, replace $\mathbb{R}$ by $L_\delta = \{x + i\delta\}$, $\delta > 0$: $f(z) = e^{i(z^3/3+sz)}$ is $O(e^{-\delta x^2})$ on L_δ and $O(e^{-yR^2})$ on the vertical sides of rectangles γ_R, so by Cauchy

$$\mathrm{Ai}(s) = \frac{1}{2\pi} \int_{L_\delta} e^{i(z^3/3+sz)} \, dz,$$

and since $f(z) = O(e^{-\delta x^2})$ on L_δ for every complex s, Ai extends to an entire function. It satisfies

$$\mathrm{Ai}''(s) = s \, \mathrm{Ai}(s) : \tag{17}$$

$\mathrm{Ai}''(s) - s \, \mathrm{Ai}(s) = \frac{1}{2\pi} \int_{L_\delta} (-z^2 - s) e^{i(z^3/3+sz)} dz = \frac{i}{2\pi} \int_{L_\delta} \frac{d}{dz}(f(z)) \, dz = 0$, since f vanishes at infinity along L_δ. Comparing (17) with $y'' = Ay$ (A large) suggests $\mathrm{Ai}(s)$ decreases rapidly as $s \to \infty$ (it stays bounded by (16), excluding $e^{\sqrt{A}s}$) and oscillates as $s \to -\infty$.

Theorem 3.1

Suppose $u > 0$. Then as $u \to \infty$, (i) $\mathrm{Ai}(-u) = \pi^{-1/2}u^{-1/4}\cos\left(\frac{2}{3}u^{3/2} - \frac{\pi}{4}\right)(1 + O(1/u^{3/4}))$.
(ii) $\mathrm{Ai}(u) = \dfrac{1}{2\pi^{1/2}}u^{-1/4}e^{-\frac{2}{3}u^{3/2}}(1 + O(1/u^{3/4}))$.

Proof. (i) Change variables $x \mapsto u^{1/2}x$ with $s = -u$: $\mathrm{Ai}(-u) = u^{1/2}I_-(u^{3/2})$, where

$$I_-(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{it(x^3/3-x)}\,dx = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-s\Phi(x)}\,dx \tag{19}$$

with $s = -it$, $\Phi(x) = x^3/3 - x$: critical points ± 1, $\Phi''(x) = 2x$, $\Phi(\pm 1) = \mp 2/3$. Split into $[-2, 0]$, $[0, 2]$ and $(-\infty, -2]$, $[2, \infty)$. On $[0, 2]$, Proposition 2.2 ($s = -it$, $x_0 = 1$, $\psi = 1/2\pi$, $\Phi''(1) = 2$) gives $\frac{1}{2\sqrt{\pi}}e^{-i\frac{2}{3}t}\left[\frac{1}{(-it)^{1/2}} + O(\frac{1}{|t|})\right]$; on $[-2, 0]$ (a maximum; use $-\Phi$, $s = it$) $\frac{1}{2\sqrt{\pi}}e^{i\frac{2}{3}t}\left[\frac{1}{(it)^{1/2}} + O(\frac{1}{|t|})\right]$. The tails are $O(1/|t|)$ by parts ($\Phi' \neq 0$), e.g. $\int_{-\infty}^{-2} e^{it\Phi}dx = \lim_N \frac{1}{it}\int_{-N}^{-2}\frac{d}{dx}(e^{it\Phi})\frac{dx}{\Phi'(x)}$. Insert into $\mathrm{Ai}(-u) = u^{1/2}I_-(u^{3/2})$: (i).

(ii) With $s = u$, $\mathrm{Ai}(u) = u^{1/2}I_+(u^{3/2})$,

$$I_+(s) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-sF(x)}\,dx, \qquad F(x) = -i(x^3/3 + x). \tag{20}$$

No real critical point exists; integration by parts shows rapid decay but not its order. Deform to a contour γ on which (a) $\mathrm{Im}(F) = 0$ and (b) $\mathrm{Re}(F)$ has a non-degenerate minimum at z_0; then $F'(z_0) = 0$, and of the two orthogonal curves through z_0 where F is real, take the one γ_1 where $\mathrm{Re}\,F$ has a minimum (on γ_2 a maximum): the **method of steepest descent** ($-\mathrm{Re}\,F$ has a saddle at z_0). Here $\mathrm{Re}(F) = x^2y - y^3/3 + y$, $\mathrm{Im}(F) = -x^3/3 + xy^2 - x$, $F'(z) = -i(z^2 + 1)$, $z_0 = \pm i$. At i, $\mathrm{Im}\,F = 0$ on $\gamma_1 = \{y^2 = x^2/3 + 1\}$ and $\gamma_2 = \{x = 0\}$ (maximum there), so γ is the branch $y = (x^2/3 + 1)^{1/2}$, asymptotic to the rays $re^{i\pi/6}$, $re^{i5\pi/6}$. Then

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-sF(x)}\,dx = \frac{1}{2\pi}\int_{\gamma} e^{-sF(z)}\,dz, \tag{21}$$

by Cauchy's theorem between $\mathbb{R}$ and γ inside the circle of radius R ($e^{-sF} = O(e^{-cyx^2})$, so the arcs are $O(\int_0^{\pi/6} e^{-cR^2\sin\theta}R\,d\theta) = O(1/R)$). On γ, $\Phi(x) = \mathrm{Re}(F) = (\frac{8}{9}x^2 + \frac{2}{3})(x^2/3 + 1)^{1/2}$, $dz = dx + i\frac{x}{3}(x^2/3 + 1)^{-1/2}dx$; Φ is even and $x(x^2/3 + 1)^{-1/2}$ odd, so

$$\frac{1}{2\pi}\int_{\gamma} e^{-sF(z)}\,dz = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-s\Phi(x)}\,dx. \tag{22}$$

As $(1 + u)^{1/2} = 1 + u/2 + O(u^2)$, $\Phi(x) = \frac{2}{3} + x^2 + O(x^4)$, so $\Phi(0) = 2/3$, $\Phi''(0) = 2$. Proposition 2.1 on $[-c, c]$ with $\psi = 1/2\pi$ gives $e^{-\frac{2}{3}s}\frac{1}{2\pi^{1/2}}\left[\frac{1}{s^{1/2}} + O(\frac{1}{s})\right]$; the rest is $O(e^{-2s/3}e^{-\delta s})$, since there $\Phi(x) \geq \frac{2}{3} + c_1x^2$. So $I_+(s) = e^{-\frac{2}{3}s}\frac{1}{2\pi^{1/2}}\left[\frac{1}{s^{1/2}} + O(\frac{1}{s})\right]$, which gives (ii). ∎

4 The partition function

Motivation. The generating function of $p(n)$ is a product of geometric series; saddle-point methods give the size of $p(n)$.

Theorem 4.1 (Hardy–Ramanujan)

If p denotes the partition function, then (i) $p(n) \sim \dfrac{1}{4\sqrt{3}\,n} e^{Kn^{1/2}}$ as $n \to \infty$, where $K = \pi\sqrt{2/3}$.
(ii) A much more precise assertion is that

$$p(n) = \frac{1}{2\pi\sqrt{2}} \frac{d}{dn}\Big(\frac{e^{K(n-\frac{1}{24})^{1/2}}}{(n-\frac{1}{24})^{1/2}}\Big) + O(e^{\frac{K}{2}n^{1/2}}).$$

(Since $(n-\frac{1}{24})^{1/2} - n^{1/2} = O(n^{-1/2})$, $e^{K(n-1/24)^{1/2}} \sim e^{Kn^{1/2}}$, and (ii) implies (i).)

Heuristic principle: if $\sum F_n w^n$ converges in the unit disc, the asymptotics of F_n are governed by the "singularities" of F on the unit circle and their "orders". (If F is meromorphic in a larger disc with only a pole of order r at $w = 1$ on the circle, $F_n = P(n) + O(e^{-\epsilon n})$ with $\deg P = r - 1$, $\sum P(n) w^n$ being essentially the principal part.)

By Euler's formula (Theorem 2.1), with $w = e^{2\pi i z}$, $z = x + iy$, $y > 0$,

$$\sum_{n=0}^{\infty} p(n) e^{2\pi i n z} = f(z) = \prod_{n=1}^{\infty} \frac{1}{1 - e^{2\pi i n z}}, \qquad p(n) = \int_\gamma f(z) e^{-2\pi i n z}\, dz, \tag{23}$$

γ the segment from $-1/2 + i\delta$ to $1/2 + i\delta$, $\delta = \delta(n) > 0$. f is largest near 0: $|f(x+iy)| \le f(iy)$ (positive coefficients), increasing as y decreases. Now $f(z) = e^{\pi i z/12}\eta(z)^{-1}$ with the Dedekind $\eta(z) = e^{\pi i z/12}\prod(1 - e^{2\pi i n z})$ (whence the 1/24), and by $\eta(-1/z) = \sqrt{z/i}\,\eta(z)$ (Proposition 1.9),

$$f(z) = \sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} f(-1/z), \tag{24}$$

and as $\mathrm{Im}(-1/z) \to \infty$, $f(-1/z) \to 1$ rapidly, since

$$f(z) = 1 + O(e^{-2\pi y}), \qquad z = x + iy,\ y \ge 1. \tag{25}$$

So $f_1(z) = \sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}}$ approximates f near 0, and by (24) $p(n) = p_1(n) + E(n)$ with

$$p_1(n) = \int_\gamma \sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} e^{-2\pi i n z}\, dz, \qquad E(n) = \int_\gamma \sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} e^{-2\pi i n z}(f(-1/z) - 1)\, dz.$$

The error term. For $z \in \gamma$, since $\mathrm{Re}(i/z) = \delta/(\delta^2 + x^2)$,

$$\Big|\sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} e^{-2\pi i n z}\Big| \le c e^{2\pi n\delta} e^{\frac{\pi}{12}\frac{\delta}{\delta^2+x^2}}. \tag{26}$$

By (25) with z replaced by $-1/z$,

$$|f(-1/z) - 1| \le c e^{-2\pi\frac{\delta}{\delta^2+x^2}} \qquad \text{if } \frac{\delta}{\delta^2 + x^2} \ge 1; \tag{27}$$

and since $|f(z)| \le f(iy) \le C e^{\frac{\pi}{12y}}$ for $y \le 1$ (by (24)),

$$|f(-1/z) - 1| = O(e^{\frac{\pi}{12}\frac{\delta^2+x^2}{\delta}}) = O(e^{\frac{\pi}{48\delta}}) \qquad \text{if } \frac{\delta}{\delta^2 + x^2} \le 1, \tag{28}$$

as $|x| \le 1/2$. Using (26), (27) when $\frac{\delta}{\delta^2+x^2} \ge 1$ gives $O(e^{2\pi n\delta})$ (as $2\pi > \pi/12$); using (26), (28) otherwise gives $O(e^{2\pi n\delta} e^{\frac{\pi}{48\delta}})$. Choose δ to minimize, $2\pi n\delta = \frac{\pi}{48\delta}$, i.e. $\delta = \frac{1}{4\sqrt{6}} n^{-1/2}$; then

$$E(n) = O(e^{\frac{\pi}{\sqrt{6}}n^{1/2}}) = O(e^{\frac{K}{2}n^{1/2}}).$$

The main term. Add to γ the end-segments $[-1/2,-1/2+i\delta]$ and $[1/2+i\delta,1/2]$, forming Γ; on them $\sqrt{z/i}\,e^{\frac{\pi i}{12z}}=O(1)$, so they contribute $O(e^{2\pi n\delta})=O(e^{\frac{K}{4}n^{1/2}})$, absorbed in $E(n)$. So

$$p_1(n)=\int_\Gamma \sqrt{z/i}\,e^{\frac{\pi i}{12z}}e^{\frac{\pi i z}{12}}e^{-2\pi i n z}\,dz. \tag{29}$$

Change variables $z\mapsto \mu z$ so that the exponentials combine to $e^{Ai(\frac{1}{z}-z)}$: this requires $A=2\pi\mu(n-\frac{1}{24})$ and $A=\frac{\pi}{12\mu}$, i.e.

$$A=\frac{\pi}{\sqrt{6}}\Big(n-\frac{1}{24}\Big)^{1/2},\qquad \mu=\frac{1}{2\sqrt{6}}\Big(n-\frac{1}{24}\Big)^{-1/2}.$$

Then

$$p_1(n)=\mu^{3/2}\int_{\Gamma'}e^{-sF(z)}\sqrt{z/i}\,dz,\qquad F(z)=i(z-1/z),\quad s=A, \tag{30}$$

$\Gamma'=\mu^{-1}\Gamma$ the union of $[-a_n,-a_n+i\delta']$, $[-a_n+i\delta',a_n+i\delta']$, $[a_n+i\delta',a_n]$, $a_n=\frac12\mu^{-1}\approx\sqrt{6}\,n^{1/2}$, $\delta'=\mu^{-1}\delta\to1$. F has the critical point i; through it F is real on the imaginary axis (maximum, rejected) and on the unit circle (minimum). By Cauchy replace Γ' by $[-a_n,-1]$, $[1,a_n]$ and the upper unit semicircle. On the segments F is purely imaginary, contributing $O(a_n^{3/2}\mu^{3/2})=O(1)$. On $z=e^{i\theta}$ (θ from π to 0), $dz=ie^{i\theta}d\theta$, $i(z-1/z)=-2\sin\theta$, $\sqrt{z/i}=e^{i(\theta/2-\pi/4)}$; with $\theta=\varphi+\pi/2$ the integrand is $-e^{2s\cos\varphi}e^{3i\varphi/2}$, contributing

$$\mu^{3/2}\int_{-\pi/2}^{\pi/2}e^{2s\cos\varphi}\big(\cos(3\varphi/2)+i\sin(3\varphi/2)\big)\,d\varphi.$$

Laplace's method with $\Phi(\varphi)=-\cos\varphi$, $\varphi_0=0$, $\Phi(0)=-1$, $\Phi''(0)=1$, $\psi(\varphi)=\cos(3\varphi/2)+i\sin(3\varphi/2)$, $\psi(0)=1$, gives $\mu^{3/2}e^{2s}\frac{\sqrt{2\pi}}{(2s)^{1/2}}(1+O(s^{-1/2}))$. Since $2s=K(n-\frac{1}{24})^{1/2}$ and $\mu=\frac{1}{2\sqrt6}(n-\frac{1}{24})^{-1/2}$,

$$p(n)=\frac{1}{4\sqrt3\,n}e^{Kn^{1/2}}(1+O(n^{-1/4})),$$

which is (i).

The precise formula (ii). Write $p_1(n)=\frac{d}{dn}q(n)+e(n)$, where

$$q(n)=\frac{1}{2\pi}\int_\Gamma (z/i)^{-1/2}e^{\frac{\pi i}{12z}}e^{\frac{\pi i z}{12}}e^{-2\pi i n z}\,dz$$

(differentiating $e^{-2\pi inz}$ in n produces $-2\pi iz$, and $\frac{-2\pi iz}{2\pi}(z/i)^{-1/2}=(z/i)^{1/2}$), and $e(n)$ comes from the dependence of $\Gamma=\Gamma_n$ on n; by Cauchy's theorem it is $O(e^{2\pi n\delta})=O(e^{\frac{K}{4}n^{1/2}})$. As before,

$$q(n)=\frac{\mu^{1/2}}{2\pi}\int_{\Gamma''}e^{-sF(z)}(z/i)^{-1/2}\,dz, \tag{31}$$

the segments contributing $O(a_n^{1/2}\mu^{1/2})=O(1)$ to $\frac{d}{dn}q(n)$. On the semicircle the integral equals

$$\frac{\mu^{1/2}}{2\pi}\int_{-\pi/2}^{\pi/2}e^{2s\cos\varphi}\big(\cos(\varphi/2)+i\sin(\varphi/2)\big)\,d\varphi=\frac{\mu^{1/2}}{2\pi}\int_{-\pi/2}^{\pi/2}e^{2s\cos\varphi}\cos(\varphi/2)\,d\varphi,$$

the odd part vanishing. With $\cos\varphi=1-2\sin^2(\varphi/2)$ and $x=\sin(\varphi/2)$, $\cos(\varphi/2)d\varphi=2dx$, this is

$$\frac{\mu^{1/2}}{\pi}e^{2s}\int_{-1/\sqrt2}^{1/\sqrt2}e^{-4sx^2}\,dx,\qquad \int_{-1/\sqrt2}^{1/\sqrt2}e^{-4sx^2}\,dx=\frac{\sqrt\pi}{2s^{1/2}}+O(e^{-2s}),$$

and the same holds after differentiating in s. Collecting the error terms,

$$p(n)=\frac{d}{dn}\Big(\frac{\mu^{1/2}e^{2s}}{2\sqrt\pi\,s^{1/2}}\Big)+O\big(e^{\frac{K}{2}n^{1/2}}\big).$$

Since $\mu^{1/2}/s^{1/2}=(2\pi)^{-1/2}(n-\frac{1}{24})^{-1/2}$ and $2s=K(n-\frac{1}{24})^{1/2}$, this is (ii).

5 Selected exercises

Motivation. Two problems from Stein–Shakarchi (numbers kept).

Exercise A.1 (Smooth cut-offs)

Let η be an indefinitely differentiable function on $\mathbb{R}$, supported in a finite interval, with $\eta(x) = 1$ for x near 0. Show that if $m > -1$ and $N > 0$, then

$$\int_0^\infty e^{-sx} x^m \eta(x)\, dx = s^{-m-1}\Gamma(m+1) + O(|s|^{-N})$$

as $|s| \to \infty$ with $\mathrm{Re}(s) \geq 0$ (principal branch of s^{-m-1}).

Solution. Fix $\delta, b > 0$ with $\eta = 1$ on $[0, \delta]$ and $\eta = 0$ on $[b, \infty)$; we may take N to be an integer. Throughout $\mathrm{Re}(s) \geq 0$, $s \neq 0$, so $|e^{-sx}| \leq 1$ for $x \geq 0$, and $e^{-sx} = -\frac{1}{s}\frac{d}{dx}e^{-sx}$.

Step 1. (Functions vanishing near 0.) If φ is indefinitely differentiable with compact support in $(0, \infty)$, then N integrations by parts (no boundary terms) give

$$\Big|\int_0^\infty e^{-sx}\varphi(x)\, dx\Big| = \Big|s^{-N}\int_0^\infty e^{-sx}\varphi^{(N)}(x)\, dx\Big| \leq \|\varphi^{(N)}\|_{L^1}|s|^{-N}.$$

Step 2. (Case $m = 0$.) One integration by parts gives

$$\int_0^\infty e^{-sx}\eta(x)\, dx = \Big[-\frac{e^{-sx}\eta(x)}{s}\Big]_0^\infty + \frac{1}{s}\int_0^\infty e^{-sx}\eta'(x)\, dx = \frac{1}{s} + O(|s|^{-N-1})$$

by Step 1 applied to $\varphi = \eta'$, which vanishes on $[0, \delta]$ and on $[b, \infty)$. Since $\Gamma(1) = 1$ this is the claim.

Step 3. (Case $-1 < m < 0$.) By the proof of Proposition 1.3,

$$\int_0^\infty e^{-sx}x^m\, dx = \lim_{R\to\infty}\int_0^R e^{-sx}x^m\, dx = s^{-m-1}\Gamma(m+1).$$

So, with $g(x) = x^m(1 - \eta(x))$, it suffices to show

$$J(s) = \lim_{R\to\infty}\int_0^R e^{-sx}g(x)\, dx = O(|s|^{-N}).$$

Now g is indefinitely differentiable on $[0, \infty)$, vanishes on $[0, \delta]$, and equals x^m on $[b, \infty)$. Hence $g^{(j)}(R) = O(R^{m-j}) \to 0$ for all $j \geq 0$, and $g^{(j)} \in L^1(0, \infty)$ for $j \geq 1$ since $m - j < -1$. One integration by parts gives

$$\int_0^R e^{-sx}g\, dx = -\frac{e^{-sR}g(R)}{s} + \frac{1}{s}\int_0^R e^{-sx}g'\, dx \longrightarrow \frac{1}{s}\int_0^\infty e^{-sx}g'(x)\, dx \quad (R \to \infty).$$

Repeating with $g', \ldots, g^{(N-1)}$ (the boundary terms at R tend to 0, those at 0 vanish), $J(s) = s^{-N}\int_0^\infty e^{-sx}g^{(N)}(x)\, dx$, so $|J(s)| \leq \|g^{(N)}\|_{L^1}|s|^{-N}$.

Step 4. (General $m > -1$.) Take $k \geq 0$ with $k - 1 < m \leq k$, $\mu = m - k \in (-1, 0]$. For $j \leq k - 1$, $(x^m)^{(j)} \propto x^{m-j}$ vanishes at 0, so k integrations by parts give

$$\int_0^\infty (x^m)^{(k)}e^{-sx}\eta\, dx = (-1)^k\int_0^\infty x^m(e^{-sx}\eta)^{(k)}dx.$$

By Leibniz, $(e^{-sx}\eta)^{(k)} = (-s)^k e^{-sx}\eta + \sum_{j=1}^k \binom{k}{j}(-s)^{k-j}e^{-sx}\eta^{(j)}$. For $j \geq 1$, $x^m\eta^{(j)}$ has compact support in $(0, \infty)$, so Step 1 (with $N + k$) makes its contribution $O(|s|^{k-j-N-k}) = O(|s|^{-N})$. With $c = m(m-1)\cdots(m-k+1)$, $(x^m)^{(k)} = c\, x^\mu$, and Steps 2–3 give

$$s^k\int_0^\infty e^{-sx}x^m\eta\, dx = c\, s^{-\mu-1}\Gamma(\mu+1) + O(|s|^{-N}).$$

Step 5. (Conclusion.) Divide by s^k: since $s^{-\mu-1}s^{-k} = s^{-m-1}$ for the principal branch and $c\,\Gamma(\mu+1) = \Gamma(m+1)$ by k applications of $\Gamma(z+1) = z\Gamma(z)$,

$$\int_0^\infty e^{-sx}x^m\eta(x)\,dx = s^{-m-1}\Gamma(m+1) + O(|s|^{-N-k}),$$

which is stronger than claimed. ∎

Exercise A.4 (Poles on the unit circle)

Let $F(w) = \sum_{n\geq 0} F_n w^n$ be meromorphic near the closed unit disc with poles there only at $\alpha_1, \dots, \alpha_k$ on the circle, of orders $r_1, \dots, r_k$. Let

$$P_j(n) = \frac{1}{(r_j-1)!}\Big(\frac{d}{dw}\Big)^{r_j-1}\big[(w-\alpha_j)^{r_j}w^{-n-1}F(w)\big]_{w=\alpha_j}.$$

(a) Show that for some $\epsilon > 0$, $F_n = -\sum_{j=1}^k P_j(n) + O(e^{-\epsilon n})$ as $n \to \infty$.
(b) Show that $P_j(n) = \alpha_j^{-n}Q_j(n)$ with Q_j a polynomial of degree $r_j - 1$.
(c) Find F_n exactly for $F(w) = \frac{1}{(1-w)(1-w^2)}$, and check it against the number of solutions of $a + 2b = n$ in integers $a, b \geq 0$.

Solution.

(a)

Step 1. (A larger circle.) $\{|w| \leq R_0\} \subset \Omega$ for some $R_0 > 1$, containing finitely many poles; choose $R \in (1, R_0)$ below the moduli of those outside the unit circle. Then F is holomorphic on $\{|w| \leq R\}$ except at the α_j.

Step 2. (Residues.) Let $G(w) = w^{-n-1}F(w)$; $\operatorname{res}_0 G = F_n$. At α_j, G has a pole of order $\leq r_j$, and the coefficient of $(w-\alpha_j)^{-1}$ is the order-(r_j-1) Taylor coefficient of $(w-\alpha_j)^{r_j}G$, so $\operatorname{res}_{\alpha_j} G = P_j(n)$. The residue theorem on $|w| = R$ gives

$$F_n + \sum_{j=1}^k P_j(n) = \frac{1}{2\pi i}\oint_{|w|=R} w^{-n-1}F(w)\,dw.$$

Step 3. (Estimate.) With $M = \max_{|w|=R}|F|$, the right side is at most $R \cdot R^{-n-1}M = MR^{-n} = Me^{-\epsilon n}$ with $\epsilon = \log R > 0$.

(The sign matters: for $F = 1/(1-w)$ one has $F_n = 1$ and $P_1(n) = -1$.)

(b) Let $h(w) = (w-\alpha_j)^{r_j}F(w)$, holomorphic near α_j with $h(\alpha_j) \neq 0$ because the pole has order exactly r_j. Since $(\frac{d}{dw})^i w^{-n-1} = (-1)^i(n+1)\cdots(n+i)\,w^{-n-1-i}$, Leibniz's rule gives

$$P_j(n) = \frac{\alpha_j^{-n}}{(r_j-1)!}\sum_{i=0}^{r_j-1}\binom{r_j-1}{i}(-1)^i(n+1)\cdots(n+i)\,\alpha_j^{-1-i}h^{(r_j-1-i)}(\alpha_j).$$

The sum is a polynomial $Q_j(n)$; only $i = r_j - 1$ contributes to degree $r_j - 1$, with leading coefficient $(-1)^{r_j-1}\alpha_j^{-r_j}h(\alpha_j)/(r_j-1)! \neq 0$. So $P_j(n) = \alpha_j^{-n}Q_j(n)$, and since $|\alpha_j| = 1$, $|P_j(n)|$ grows like n^{r_j-1}.

(c)

Step 1. $F(w) = \frac{1}{(1-w)^2(1+w)}$ is rational with poles 1 (order 2) and -1 (order 1). Here $w^{-n-1}F(w) = O(R^{-n-4})$ on $|w| = R$, so the integral in (a) tends to 0 as $R \to \infty$, and $F_n = -P_1(n) - P_2(n)$ exactly.

Step 2. At $\alpha = -1$: $P_2(n) = \left[\frac{w^{-n-1}}{(1-w)^2}\right]_{w=-1} = \frac{(-1)^{n+1}}{4}$.

Step 3. At $\alpha = 1$: $(w-1)^2 w^{-n-1} F(w) = \frac{w^{-n-1}}{1+w}$, so

$$P_1(n) = \Big[\frac{-(n+1)w^{-n-2}}{1+w} - \frac{w^{-n-1}}{(1+w)^2}\Big]_{w=1} = -\frac{n+1}{2} - \frac{1}{4} = -\frac{2n+3}{4}.$$

Step 4. Hence $F_n = \frac{2n+3}{4} + \frac{(-1)^n}{4}$, i.e. $F_n = \frac{n}{2} + 1$ for even n and $\frac{n+1}{2}$ for odd n; in both cases $F_n = \lfloor n/2 \rfloor + 1$. On the other hand $F(w) = \sum_{a \geq 0} w^a \sum_{b \geq 0} w^{2b}$, so F_n counts the pairs (a, b) with $a + 2b = n$, i.e. $b = 0, 1, \ldots, \lfloor n/2 \rfloor$: again $\lfloor n/2 \rfloor + 1$. ■

Appendix B. Simple Connectivity and Jordan Curve Theorem

Overview

Recall (Chapter 3) that Ω is *simply connected* if any two curves in Ω with the same end-points are homotopic; then

$$\int_\eta f(\zeta)\,d\zeta = 0 \tag{1}$$

for every closed curve $\eta \subset \Omega$ and every f holomorphic in Ω. In this appendix we prove:

(I) Ω is simply connected if and only if it is *holomorphically simply connected*, i.e. (1) holds for all closed $\eta \subset \Omega$ and all f holomorphic in Ω.

(II) A bounded region is simply connected if and only if its complement is connected.

(III) Ω is simply connected if and only if no curve in Ω winds around a point of Ω^c.

Second, what region does a simple closed curve γ "enclose"? We prove the Jordan curve theorem for piecewise-smooth curves (γ^c has two components, the bounded one simply connected, both with boundary γ) and a general Cauchy theorem extending the toy-contour version of Chapter 2. "Curve" means "piecewise-smooth curve".

1 Equivalent formulations of simple connectivity

Motivation. Simple connectivity, vanishing integrals, winding numbers and connected complements are equivalent tests.

Theorem 1.1

A region Ω is holomorphically simply connected if and only if Ω is simply connected.

Proof. If Ω is simply connected it is holomorphically simply connected (Chapter 3, Corollary 5.3). Conversely, if $\Omega \neq \mathbb{C}$ is holomorphically simply connected, the proof of the Riemann mapping theorem (Theorem 3.1) applies verbatim, since it uses simple connectivity only for a logarithm (and square root) $\log g(z) = c + \int_{z_0}^{z} g'/g$ of a non-vanishing g, well defined by (1). So Ω is conformally equivalent to $\mathbb{D}$, hence simply connected (conformal maps carry homotopies to homotopies). ∎

Theorem 1.2

If Ω is a bounded region in $\mathbb{C}$, then Ω is simply connected if and only if Ω^c is connected.

Boundedness is needed: a strip is simply connected with disconnected complement. With complements in the Riemann sphere the conclusion holds for all regions.

Proof that Ω^c connected $\Rightarrow$ Ω simply connected. By Theorem 1.1 it suffices to show $\int_\eta f = 0$ for $\eta \subset \Omega$ closed and f holomorphic in Ω. With $d(z, \Omega^c) = \inf\{|z - w| : w \in \Omega^c\}$, let

$$K_\delta = \{z \in \overline{\Omega} : d(z, \Omega^c) \geq \delta\}.$$

K_δ is compact (Ω is bounded), $K_\delta \subset \Omega$, and $\eta \subset K_\delta$ for small δ since η is compact. We show that K_δ^c is connected.

Suppose $K_\delta^c = O_1 \cup O_2$ with O_1, O_2 disjoint, open, non-empty, and set $F_i = O_i \cap \Omega^c$. Since $\Omega^c \subset K_\delta^c$, $\Omega^c = F_1 \cup F_2$ with $F_1 \cap F_2 = \emptyset$.

Step F_1 is closed. Let $z_n \in F_1$, $z_n \to z$. Then $z \in \Omega^c$ (Ω^c is closed), so $z \in O_1 \cup O_2$. If $z \in O_2$, then $z_n \in O_2$ for large n as O_2 is open, contradicting $z_n \in O_1$. So $z \in O_1$, i.e. $z \in F_1$. Likewise F_2 is closed.

Step $F_1 \neq \emptyset$. Otherwise $\Omega^c \subset O_2$. For $w \in O_1$ ($w \notin K_\delta$) there is $z \in \Omega^c$ with $|w - z| < \delta$; the segment $[w, z]$ lies in $K_\delta^c = O_1 \cup O_2$ and meets both, a contradiction. Likewise $F_2 \neq \emptyset$.

So Ω^c would be disconnected; hence K_δ^c is connected, and Runge (Chapter 2, Theorem 5.7) gives polynomials $P_n \to f$ uniformly on $K_\delta \supset \eta$, so $\int_\eta f = \lim \int_\eta P_n = 0$. ∎

The converse is proved after Theorem 1.4, using winding numbers.

Winding numbers

If γ is closed and $z \notin \gamma$, each loop of γ around z changes $\frac{1}{2\pi} \arg(\zeta - z)$ by ± 1. Since $\log w = \log|w| + i \arg w$, the quantity $\frac{1}{2\pi i} \int_\gamma \frac{d\zeta}{\zeta - z}$ should count these loops.

Definition (Winding number)

The **winding number** (or **index**) of a closed curve γ around $z \notin \gamma$ is

$$W_\gamma(z) = \frac{1}{2\pi i} \int_\gamma \frac{d\zeta}{\zeta - z}.$$

For $\gamma(t) = e^{ikt}$, $0 \le t \le 2\pi$ ($k \in \mathbb{N}$), $W_\gamma(z) = k$ for $|z| < 1$ and 0 for $|z| > 1$; for e^{-ikt} the values are $-k$ and 0. For a positively oriented toy contour, $W_\gamma = 1$ in its interior and 0 in its exterior.

Lemma 1.3

Let γ be a closed curve in $\mathbb{C}$.

(i) If $z \notin \gamma$, then $W_\gamma(z) \in \mathbb{Z}$.

(ii) If z, w lie in the same connected component of γ^c, then $W_\gamma(z) = W_\gamma(w)$.

(iii) If z lies in the unbounded component of γ^c, then $W_\gamma(z) = 0$.

Proof. (i) Let $G(t) = \int_0^t \frac{\gamma'(s)}{\gamma(s) - z}\, ds$, continuous and, off finitely many points, differentiable with $G' = \gamma'/(\gamma - z)$. Then $H = (\gamma - z)e^{-G}$ has $H' = 0$ off finitely many points, so $H(1) = H(0)$; as $\gamma(1) = \gamma(0) \neq z$, $e^{-G(1)} = 1$ and $G(1) = 2\pi i\, W_\gamma(z) \in 2\pi i\mathbb{Z}$.

(ii) W_γ is a continuous function of $z \in \gamma^c$ (the integrand depends continuously on z, uniformly on γ), integer-valued by (i), hence constant on each connected component.

(iii) $|W_\gamma(z)| \le \frac{\ell(\gamma)}{2\pi\, d(z,\gamma)} \to 0$ as $|z| \to \infty$; since W_γ is constant on the unbounded component by (ii), it is 0 there. ∎

Theorem 1.4

A bounded region Ω is simply connected if and only if $W_\gamma(z) = 0$ for every closed curve $\gamma \subset \Omega$ and every $z \notin \Omega$.

Proof. If Ω is simply connected and $z \notin \Omega$, then $f(\zeta) = 1/(\zeta - z)$ is holomorphic in Ω, and Cauchy's theorem gives $W_\gamma(z) = 0$.

Conversely, by the part of Theorem 1.2 already proved, it suffices that Ω^c be connected. Otherwise $\Omega^c = F_1 \cup F_2$, disjoint closed non-empty, and only one is unbounded ($\Omega^c \supset \{|z| > R\}$), say F_2; so F_1 is compact. We find a closed $\gamma \subset \Omega$ and $w \in F_1$ with $W_\gamma(w) \neq 0$, using: ■

Lemma 1.5

Under the above assumptions, let $w \in F_1$. There is a finite collection $\mathcal{Q} = \{Q_1, \dots, Q_n\}$ of closed squares of a uniform grid $\mathcal{G}$ such that:

(i) w is in the interior of Q_1;

(ii) the interiors of Q_j and Q_k are disjoint for $j \neq k$;

(iii) F_1 is contained in the interior of $\bigcup_j Q_j$;

(iv) $\bigcup_j Q_j$ is disjoint from F_2;

(v) the boundary of $\bigcup_j Q_j$ lies in Ω and is a finite union of disjoint simple closed polygonal curves.

Proof of Theorem 1.4, assuming Lemma 1.5. Orient each ∂Q_j positively. Since w is interior to Q_1 and $w \notin Q_j$ for $j > 1$,

$$\sum_{j=1}^{n} \frac{1}{2\pi i} \int_{\partial Q_j} \frac{d\zeta}{\zeta - w} = 1. \tag{2}$$

Shared sides cancel; the remaining sides, with $\bigcup Q_j$ on the left, form the polygons $\gamma_1, \dots, \gamma_M$ of (v), so $\sum_j W_{\gamma_j}(w) = 1$ and some $W_{\gamma_{j_0}}(w) \neq 0$ with $\gamma_{j_0} \subset \Omega$, $w \notin \Omega$: a contradiction. ■

Proof of Lemma 1.5. $d = d(F_1, F_2) > 0$. Take a grid $\mathcal{G}_0$ of closed squares of side $\delta < d/100$ with w the center of a square R_1, and let $\mathcal{R} = \{R_1, \dots, R_m\}$ be those meeting F_1.

Step (i)–(iv) and $\partial \bigcup R_j \subset \Omega$. (i) and (ii) are clear. Points of $\bigcup R_j$ are within $\sqrt{2}\delta < d$ of F_1: (iv). Every grid square containing $z \in F_1$ is in $\mathcal{R}$, and these cover a neighborhood of z: (iii). A *boundary side* (side of exactly one square of $\mathcal{R}$) is a side of a square outside $\mathcal{R}$, so it misses F_1, and F_2 by (iv). So $\partial \bigcup R_j \subset \Omega$.

Step Bad vertices. End-points of boundary sides are boundary vertices. At a grid vertex P there are 0, 2 or 4 boundary sides, 4 exactly when two diagonally opposite squares are in $\mathcal{R}$ and the other two are not; such P is *bad*.

Step Refinement. Cut each square of $\mathcal{G}_0$ into 9 subsquares (grid $\mathcal{G}$), take the $9m$ subsquares of $\mathcal{R}$, Q_1 the middle one of R_1 (centered at w), and at each bad vertex P add the two subsquares with corner P in the two missing big squares; call the result $\mathcal{Q} = \{Q_1, \dots, Q_n\}$. All four small squares at P are now in $\mathcal{Q}$, so P is interior. Of the other corners of an added subsquare s, the two on sides of $\mathcal{G}_0$ have three surrounding small squares in $\mathcal{Q}$ and the inner one only s; added squares at different bad vertices do not touch. So $\mathcal{Q}$ has no bad vertices. (i)–(iv) persist (added squares are within $3\delta < d$ of F_1), and new boundary sides lie in big squares outside $\mathcal{R}$, missing F_1 and F_2; so $\partial \bigcup Q_j \subset \Omega$.

Step (v). Each boundary vertex now meets exactly two boundary sides, one entering and one leaving (orientation with $\bigcup Q_j$ on the left; check one, two adjacent or three squares). From $[a_1, a_2]$ continue uniquely $[a_2, a_3], \dots$; let $a_m = a_k$, $k < m$, be the first repetition. If $k > 1$, a_k meets three sides $[a_{k-1}, a_k]$, $[a_k, a_{k+1}]$, $[a_{m-1}, a_m]$, impossible; so $a_1 \cdots a_m$ is a simple closed polygon. Removing it and repeating gives finitely many disjoint simple closed polygons (each vertex lies on one). ■

Proof of Theorem 1.2, converse. If Ω is bounded and simply connected with disconnected complement, the proof of Theorem 1.4 gives a closed $\gamma \subset \Omega$ and $w \notin \Omega$ with $W_\gamma(w) \neq 0$, contradicting its first part. ■

2 The Jordan curve theorem

Motivation. The Jordan curve theorem guarantees that "the interior of a curve" exists; it yields a general Cauchy theorem.

The curves in the statements are piecewise-smooth, but the proofs also use curves that are merely continuous (the curves γ_η below).

Theorem 2.1 (Jordan arc theorem)

Let γ be a simple (non-closed) piecewise-smooth curve in the plane. Then γ^c is an open connected set whose boundary is precisely γ.

Theorem 2.2 (Jordan curve theorem)

For a simple closed piecewise-smooth γ, γ^c consists of two disjoint connected open sets: the bounded, simply connected **interior** Ω_γ and the unbounded **exterior** $\mathcal{U}_\gamma$. With the appropriate orientation of γ,

$$W_\gamma(z) = \begin{cases} 1 & z \in \Omega_\gamma, \\ 0 & z \in \mathcal{U}_\gamma. \end{cases}$$

Remark

Both theorems hold for merely continuous curves, but the proofs are harder; the piecewise-smooth case suffices for many applications.

Theorem 2.3

If f is holomorphic in the interior Ω_γ of a simple closed curve γ, then $\int_\eta f(\zeta)\,d\zeta = 0$ for every closed curve $\eta \subset \Omega_\gamma$.

Proof [Supplement]. Ω_γ is a simply connected region by Theorem 2.2, so this is Cauchy's theorem, Corollary 5.3 of Chapter 3. ∎

Idea of the proof of Theorem 2.1. As γ^c is open, show it is pathwise connected: join $z, w \notin \gamma$ by a curve; if it meets γ, go from z and w to points z', w' near γ, then travel "parallel" to γ, around an end-point if necessary. On a smooth piece, $i\gamma'(t)$ is normal and $\gamma(t) + i\eta\gamma'(t)$ is a parallel curve.

In the next three lemmas and two propositions, γ_0 is a *simple smooth* curve with arc-length parametrization $\gamma : [0, L] \to \mathbb{C}$, $|\gamma'| = 1$; its *interior points* are the $\gamma(t)$, $0 < t < L$ (unrelated to Theorem 2.2).

Lemma 2.4

For $\eta \in \mathbb{R}$ let γ_η be the continuous curve $\gamma_\eta(t) = \gamma(t) + i\eta\gamma'(t)$, $0 \le t \le L$. There is $\eta_1 > 0$ such that $\gamma_0 \cap \gamma_\eta = \emptyset$ whenever $0 < |\eta| < \eta_1$.

Proof.

Step Local version. For $s, t \in [0, L]$,

$$\gamma_\eta(t) - \gamma(s) = \int_s^t \gamma'(u)\,du + i\eta\gamma'(t) = \int_s^t [\gamma'(u) - \gamma'(t)]\,du + (t - s + i\eta)\gamma'(t).$$

By uniform continuity of γ' there is $\delta > 0$ with $|\gamma'(x) - \gamma'(y)| < 1/2$ for $|x - y| < \delta$. If $|s - t| < \delta$, then using $|\gamma'(t)| = 1$ and $2|a + ib| \geq |a| + |b|$,

$$|\gamma_\eta(t) - \gamma(s)| \geq |t - s + i\eta| - \frac{|t - s|}{2} \geq \frac{|\eta|}{2}.$$

So $\gamma_\eta(t) \neq \gamma(s)$ when $|t - s| < \delta$ and $\eta \neq 0$.

Step Global version. Choose $0 = t_0 < \cdots < t_n = L$ with $t_{k+1} - t_k < \delta/2$, and set $I_k = \{t : |t - t_k| \leq \delta/4\}$, $J_k = \{t : |t - t_k| < \delta/2\}$, $J_k' = \{t \in [0, L] : |t - t_k| \geq \delta/2\}$ (intersected with $[0, L]$). The I_k cover $[0, L]$. If $t \in I_k$ and $s \in J_k$ then $|t - s| < \delta$, so

$$\gamma_\eta(I_k) \cap \gamma(J_k) = \emptyset \qquad (\eta \neq 0). \tag{3}$$

Since γ_0 is simple, the compact sets $\gamma(I_k)$ and $\gamma(J_k')$ are disjoint, at distance $d_k > 0$. For $t \in I_k$, $s \in J_k'$,

$$|\gamma_\eta(t) - \gamma(s)| \geq |\gamma(t) - \gamma(s)| - |\eta| \geq d_k - |\eta| \geq d_k/2 \quad \text{if } |\eta| \leq d_k/2,$$

so

$$\gamma_\eta(I_k) \cap \gamma(J_k') = \emptyset \qquad (|\eta| < d_k/2). \tag{4}$$

With $\eta_1 = \min_k d_k/2$, (3) and (4) give $\gamma_0 \cap \gamma_\eta = \emptyset$ for $0 < |\eta| < \eta_1$. ■

Lemma 2.5

Let $z \notin \gamma_0$ and let $z_0 = \gamma(t_0)$, $0 < t_0 < L$, be a closest point of γ_0 to z (so if z is closer to an interior point than to the end-points). Then $z = \gamma(t_0) + i\eta\gamma'(t_0)$, $\eta \neq 0$ real, i.e. $z \in \gamma_\eta$.

Proof. Near t_0, $z - \gamma(t) = z - \gamma(t_0) - \gamma'(t_0)(t - t_0) + o(|t - t_0|)$, so

$$|z - z_0|^2 \leq |z - \gamma(t)|^2 = |z - z_0|^2 - 2(t - t_0)\,\mathrm{Re}\,[(z - \gamma(t_0))\overline{\gamma'(t_0)}] + o(|t - t_0|).$$

Since $t - t_0$ takes both signs, $\mathrm{Re}[(z - \gamma(t_0))\overline{\gamma'(t_0)}] = 0$, i.e. $(z - \gamma(t_0))\overline{\gamma'(t_0)} = i\eta$ with $\eta \in \mathbb{R}$. As $|\gamma'(t_0)| = 1$, $\overline{\gamma'(t_0)} = 1/\gamma'(t_0)$, so $z - \gamma(t_0) = i\eta\gamma'(t_0)$; and $\eta \neq 0$ because $z \notin \gamma_0$. ■

For $z \in \gamma_\eta$, $w \in \gamma_\epsilon$ as in Lemma 2.5, z, w are on the **same side** of γ_0 if η, ϵ have the same sign, on **opposite sides** otherwise. This refers to the chosen representations; we do not claim exclusivity. Same-side points are joined along a parallel curve; opposite sides require going around an end-point.

Proposition 2.6

Let A, B be the end-points of γ_0 and K compact with $\gamma_0 \cap K \subset \{A, B\}$. If $z, w \notin \gamma_0$ are on the same side, each closer to an interior point of γ_0 than to K or to A, B, they can be joined by a continuous curve in $(K \cup \gamma_0)^c$.

Proof. By Lemma 2.5, with $z_0 = \gamma(t_0)$, $w_0 = \gamma(s_0)$ interior points closest to z, w,

$$z = \gamma(t_0) + i\eta_0\gamma'(t_0), \qquad w = \gamma(s_0) + i\epsilon_0\gamma'(s_0),$$

with η_0, ϵ_0 of the same sign, say positive, and say $t_0 \leq s_0$.

Step Normal segments. The half-open segment from z to z_0 lies in $(K \cup \gamma_0)^c$: a point $p \neq z_0$ of it in $K \cup \gamma_0$ would have $|z - p| < d(z, K \cup \gamma_0)$. So for small $\eta > 0$, z joins $z_\eta = \gamma(t_0) + i\eta\gamma'(t_0)$, and w joins $w_\eta = \gamma(s_0) + i\eta\gamma'(s_0)$.

Step Parallel curve. Take $\eta < \eta_1$ (Lemma 2.4) and below the (positive) distance from K to $\{\gamma(t) : t_0 \leq t \leq s_0\}$, an arc avoiding A, B. Then $\{\gamma_\eta(t) : t_0 \leq t \leq s_0\}$ joins z_η to w_η, missing γ_0 (Lemma 2.4) and K ($|\gamma_\eta(t) - \gamma(t)| = \eta$). ■

Lemma 2.7

There is $\eta_2 > 0$ such that the half-neighborhood

$$N = \{\gamma(L) + \eta e^{i\theta}\gamma'(L) : -\pi/2 \le \theta \le \pi/2,\ 0 < \eta < \eta_2\}$$

is disjoint from γ_0.

Proof. As in Lemma 2.4,

$$\gamma(L) + \eta e^{i\theta}\gamma'(L) - \gamma(t) = \int_t^L [\gamma'(u) - \gamma'(L)]\,du + (L - t + \eta e^{i\theta})\gamma'(L).$$

Choose δ with $|\gamma'(u) - \gamma'(L)| < 1/2$ for $|u - L| < \delta$. Since $\cos\theta \ge 0$, $|L - t + \eta e^{i\theta}|^2 \ge (L-t)^2 + \eta^2 \ge \frac{1}{2}(L - t + \eta)^2$. So for $L - \delta < t \le L$,

$$|\gamma(L) + \eta e^{i\theta}\gamma'(L) - \gamma(t)| \ge \frac{L - t + \eta}{\sqrt{2}} - \frac{L - t}{2} \ge \frac{\eta}{\sqrt{2}} > 0.$$

Thus $\gamma(t) \notin N$ for $L - \delta < t \le L$. Finally, take η_2 less than the distance from $\gamma(L)$ to $\{\gamma(t) : 0 \le t \le L - \delta\}$, which is positive since γ_0 is simple. ∎

Proposition 2.8

Let A be an end-point of γ_0 and K compact with $\gamma_0 \cap K \subset \{A\}$. If $z, w \notin \gamma_0$ are closer to interior points of γ_0 than to K or to the end-points, they can be joined by a continuous curve in $(\gamma_0 \cup K)^c$.

Proof. For the same side use Proposition 2.6. For opposite sides, with (after reversal) $A = \gamma(0)$ and $B = \gamma(L) \notin K$, write

$$z = \gamma(t_0) + i\eta_0\gamma'(t_0), \qquad w = \gamma(s_0) - i\epsilon_0\gamma'(s_0), \qquad \eta_0, \epsilon_0 > 0,$$

and $\tau = \min(t_0, s_0) > 0$. Take $\eta > 0$ below η_1, η_2 (Lemmas 2.4, 2.7) and the (positive) distance from K to $\{\gamma(t) : \tau \le t \le L\}$. Join z to $\gamma_\eta(t_0)$ by a normal segment, follow γ_η to $\gamma(L) + i\eta\gamma'(L)$, the half-circle $\gamma(L) + \eta e^{i\theta}\gamma'(L)$, θ from $\pi/2$ to $-\pi/2$, inside N, to $\gamma(L) - i\eta\gamma'(L)$, then $\gamma_{-\eta}$ back to s_0 and the normal segment to w. The parallel pieces and the half-circle miss γ_0 (Lemmas 2.4, 2.7); all pieces are within η of $\{\gamma(t) : \tau \le t \le L\}$, so they miss K. ∎

Proof of Theorem 2.1.

Step Boundary. $O = \gamma^c$ is open, $\partial O \subset \gamma$. Each smooth point $\gamma(t)$ is a limit of points $\gamma(t) + i\eta\gamma'(t) \in O$ (Lemma 2.4 on a small sub-arc; the rest of γ is at positive distance), and smooth points are dense, so $\partial O = \gamma$.

Step Connectedness: induction on the number of smooth pieces. For smooth γ with end-points A, B and $Z, W \notin \gamma$, join Z to W by a polygonal Λ avoiding A, B. If Λ meets γ, its first and last meeting points p, q are interior points. Points $z \in \Lambda$ just before p have $d(z, \gamma) \le |z - p| \to 0$ while $|z - A|, |z - B|$ stay bounded below; take such $z \notin \gamma$ closer to an interior point than to A, B, and similarly w just after q. Proposition 2.8 with $K = \emptyset$ joins z to w in γ^c.

For n pieces, write $\gamma = K \cup \gamma_0$, K the union of $n - 1$ consecutive pieces (compact) and γ_0 smooth, $K \cap \gamma_0 = \{A\}$. By induction $Z, W \notin \gamma$ are joined in the open connected K^c by a polygonal curve avoiding the other end-point of γ_0, meeting γ_0 only at interior points. Take z (w) just before the first (after the last) meeting point, closer to an interior point of γ_0 than to K or the end-points; Proposition 2.8 joins them in $(\gamma_0 \cup K)^c = \gamma^c$. ∎

Proof of Theorem 2.2.

Step At most two components. Fix W outside a large disc containing γ; let $\mathcal{U}$ be the set of points joined to W in γ^c, open, connected, containing the exterior of the disc, and $\Omega = \gamma^c - \mathcal{U}$. Remove a smooth piece γ_0 (end-points A, B), leaving the arc K, $K \cap \gamma_0 = \{A, B\}$. For $Z \in \Omega$, Theorem 2.1 for K gives Λ_Z from Z to W missing K; it meets γ_0 (as $Z \notin \mathcal{U}$), at interior points. Choose $z, w \in \Lambda_Z$ as before, with the pieces $Z \to z$ and $w \to W$ in γ^c. Then z, w are on opposite sides (else Proposition 2.6 joins Z to W). For another $Z_1 \in \Omega$ with z_1, w_1, z and z_1 are on the same side (else z_1 is on w's side and Z_1 joins W), so Proposition 2.6 joins z to z_1 in γ^c, and Z, Z_1 are in one component. So the components are $\mathcal{U}$ and (if non-empty) Ω.

Step Exactly two components. By Lemma 1.3 it suffices to find points with different winding numbers. Fix $z_0 = \gamma(t_0)$ interior to a smooth piece, $v = \gamma'(t_0) \neq 0$, and for $\epsilon > 0$ let

$$z_\epsilon = \gamma(t_0) + i\epsilon v, \qquad w_\epsilon = \gamma(t_0) - i\epsilon v.$$

For small ϵ these avoid γ, and the normal segments $\{\gamma(t_0) \pm isv : 0 < s \leq \epsilon\}$ lie in γ^c (Lemma 2.4, plus the distance from $\gamma(t_0)$ to the rest of γ). By Lemma 1.3(ii),

$$\Delta = |W_\gamma(z_\epsilon) - W_\gamma(w_\epsilon)|$$

is a constant integer for all small $\epsilon > 0$. With $a(t) = \gamma(t) - \gamma(t_0)$,

$$\frac{\gamma'(t)}{\gamma(t) - z_\epsilon} - \frac{\gamma'(t)}{\gamma(t) - w_\epsilon} = \frac{2i\epsilon v\, \gamma'(t)}{a(t)^2 + \epsilon^2 v^2}.$$

Since $a(t) = v(t - t_0) + o(|t - t_0|)$ and $\gamma'(t) = v + o(1)$,

$$\frac{2i\epsilon v \gamma'(t)}{a(t)^2 + \epsilon^2 v^2} = \frac{2i\epsilon}{(t - t_0)^2 + \epsilon^2}\,(1 + \rho(t, \epsilon)),$$

where for $\sigma > 0$ some $\delta > 0$ gives $|\rho(t, \epsilon)| < \sigma$ for $|t - t_0| \leq \delta$, uniformly in ϵ (as $a^2 + \epsilon^2 v^2 = v^2[(t - t_0)^2 + \epsilon^2] + o((t - t_0)^2)$ and $o((t - t_0)^2)/[(t - t_0)^2 + \epsilon^2] \to 0$). Split $W_\gamma(z_\epsilon) - W_\gamma(w_\epsilon)$ at $|t - t_0| = \delta$: on $|t - t_0| \geq \delta$, $\gamma(t)$ stays away from $\gamma(t_0)$ while $z_\epsilon, w_\epsilon \to \gamma(t_0)$, so this part $\to 0$. On the rest,

$$\frac{1}{2\pi i} \int_{|t - t_0| < \delta} \frac{2i\epsilon\, dt}{(t - t_0)^2 + \epsilon^2} = \frac{1}{\pi} \int_{-\delta/\epsilon}^{\delta/\epsilon} \frac{ds}{s^2 + 1} = \frac{2}{\pi} \arctan(\delta/\epsilon) \to 1,$$

and the ρ-term is at most $\frac{\sigma}{\pi} \int_{\mathbb{R}} \frac{ds}{s^2+1} = \sigma$ in absolute value. Letting $\epsilon \to 0$ gives $|\Delta - 1| \leq \sigma$ for every $\sigma > 0$, so $\Delta = 1$.

So z_ϵ, w_ϵ lie in different components, $\mathcal{U}$ (unbounded, $W_\gamma = 0$, Lemma 1.3(iii)) and the bounded Ω, where $W_\gamma = \pm 1$ ($\Delta = 1$), $= 1$ after possibly reversing orientation. Every smooth point is a limit of z_ϵ's and w_ϵ's, smooth points are dense, and component boundaries lie in γ; so $\gamma = \partial\Omega = \partial\mathcal{U}$. Write $\Omega_\gamma = \Omega$, $\mathcal{U}_\gamma = \mathcal{U}$.

Step Ω_γ is simply connected. By Theorem 1.2 it suffices that $\Omega_\gamma^c = \gamma \cup \mathcal{U}_\gamma$ be connected. Suppose $\Omega_\gamma^c = F_1 \cup F_2$, closed, disjoint, non-empty; then $\mathcal{U}_\gamma = O_1 \cup O_2$, $O_i = \mathcal{U}_\gamma \cap F_i$ disjoint.

O_1 is open: small balls about $z \in O_1$ lie in $\mathcal{U}_\gamma \subset F_1 \cup F_2$, and if all met F_2 then $z \in F_2$; so one lies in O_1. Likewise O_2.

$O_1 \neq \emptyset$. Otherwise $F_1 \subset \gamma$ and $\mathcal{U}_\gamma \subset F_2$. Take $z \in F_1 \subset \gamma$. Since $\gamma = \partial\mathcal{U}_\gamma$, every ball at z meets $\mathcal{U}_\gamma \subset F_2$, so $z \in F_2$, a contradiction. Likewise $O_2 \neq \emptyset$.

This disconnects $\mathcal{U}_\gamma$, a contradiction. This completes the proof of the Jordan curve theorem for piecewise-smooth curves. ■

2.1 Proof of a general form of Cauchy's theorem

Theorem 2.9

If f is holomorphic in an open set containing a simple closed piecewise-smooth curve γ and its interior, then $\int_\gamma f = 0$.

Let f be holomorphic on an open $\mathcal{O} \supset \gamma \cup \Omega_\gamma$. Build a closed $\Gamma \subset \Omega_\gamma$ so close to γ that $\int_\gamma f = \int_\Gamma f$, which is 0 by Theorem 2.3. Near smooth parts Γ is essentially γ_η (Lemma 2.4); near junctions it follows circular arcs.

Lemma 2.10

Let $\gamma : [0,1] \to \mathbb{C}$ be a simple smooth curve. For all sufficiently small $\epsilon > 0$ the circle C_ϵ of center $\gamma(0)$ and radius ϵ meets γ in exactly one point.

Proof. Assume $\gamma(0) = 0$. For $\epsilon < |\gamma(1)|$ the intermediate value theorem gives an intersection. If $|\gamma(t)| = \epsilon_j$ had two solutions for $\epsilon_j \to 0$, both would tend to 0 ($|\gamma(t)| \geq c(\tau) > 0$ for $t \geq \tau$, simplicity), and the mean value theorem for $h = |\gamma|^2$ gives $t_j \to 0^+$ with $h'(t_j) = 2\,\mathrm{Re}[\gamma'(t_j)\overline{\gamma(t_j)}] = 0$. But $\gamma(t) = \gamma'(0)t + t\varphi(t)$, $\gamma'(t) = \gamma'(0) + \psi(t)$, $\varphi, \psi \to 0$, so

$$\mathrm{Re}[\gamma'(t)\overline{\gamma(t)}] = |\gamma'(0)|^2 t + o(t),$$

which is $\neq 0$ for small $t > 0$ since $\gamma'(0) \neq 0$. This is a contradiction. ∎

Proof of Theorem 2.9. Choose ϵ with $U = \{d(z,\gamma) < \epsilon\} \subset \mathcal{O}$. Let $P_1, \ldots, P_n$ be the consecutive junction points, and $\delta < \epsilon/10$ so small that the circle C_j of radius δ about P_j meets γ in exactly two points (Lemma 2.10 for the two parts at P_j; the rest of γ is farther away). Exactly one of the two arcs, $\mathcal{C}_j$, has interior in Ω_γ: for small η, $\gamma_{\pm\eta}$ near P_j lie in different components (proof of Theorem 2.2) and both meet C_j. The disc D_j of radius 2δ about P_j lies in $U \subset \mathcal{O}$.

Let B_j be the end-point of $\mathcal{C}_j$ on the part γ_j from P_j to P_{j+1}, and A_{j+1} that of $\mathcal{C}_{j+1}$ on γ_j. Take a chain of discs $\mathcal{D} = \{D_0, \ldots, D_K\}$ in U covering γ, $D_k \cap D_{k+1} \neq \emptyset$, $D_0 = D_K$, including the D_j. By Lemma 2.4 a curve Γ_j in $\Omega_\gamma \cap \bigcup \mathcal{D}$ runs parallel to γ_j from near B_j on $\mathcal{C}_j$ to near A_{j+1} on $\mathcal{C}_{j+1}$ (replaced by a nearby polygonal line, as γ is only C^1). Joining the Γ_j along pieces of the $\mathcal{C}_j$ gives a closed piecewise-smooth $\Gamma \subset \Omega_\gamma$.

f has a primitive on each disc of $\mathcal{D}$ (Chapter 2, Theorem 2.1); as in Chapter 3, Theorem 5.1, the primitives on consecutive discs differ by constants and the endpoint contributions telescope, so $\int_\gamma f = \int_\Gamma f = 0$ (Theorem 2.3). ∎